

BSCS FINAL PROJECT

ReMIND: Generative AI Based Memory Rebuilder



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ReMIND: Generative AI Based Memory Rebuilder

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Abstract

ReMIND is an AI-assisted memory reconstruction platform developed to support individuals experiencing memory loss disorders such as Alzheimer's disease, dementia, and age-related cognitive decline. The inspiration behind this project originated from a personal experience within the development team, where a family member's struggle with Alzheimer's disease highlighted the emotional, cognitive, and social challenges caused by memory deterioration. Witnessing the impact of forgotten experiences, weakened personal connections, and the loss of identity motivated the development of a technological solution capable of preserving and reconstructing meaningful memories. Memory loss remains a significant healthcare challenge, and existing digital solutions primarily rely on manual logging, reminders, or prior to recollection, making them ineffective when sufficient memory details are unavailable.

The problem addressed by ReMIND is the inability of conventional systems to assist patients in reconstructing fragmented memories from incomplete multimedia information. The project focuses on providing an intelligent, user-friendly platform that enables users to rebuild personal experiences using text notes, photographs, and voice recordings while supporting cognitive engagement and emotional well-being. The scope of the work is limited to AI-assisted memory reconstruction and contextual recall rather than medical diagnosis or clinical treatment.

To address this problem, a complete web-based platform was designed and implemented using a client-server architecture consisting of a React frontend, FastAPI backend, PostgreSQL database, Cloudinary media storage, and multiple artificial intelligence modules. The system integrates multimodal processing techniques, including natural language understanding through Groq Llama 3, speech transcription using ElevenLabs Speech-to-Text, image enhancement through GFPGAN and Real-ESRGAN, image caption generation using BLIP, and voice synthesis through Google Text-to-Speech. These components collaboratively analyze multimedia inputs, extract contextual information, generate coherent memory narratives, and present reconstructed memories through an interactive chat-based interface. Additional functionality includes secure authentication, Google OAuth integration, memory timelines, reminder management, administrative controls, evaluation of metrics, and cloud deployment.

The completed prototype successfully demonstrates end-to-end memory reconstruction from multimodal inputs and supports text, image, and voice-based interaction within a unified platform. System testing confirmed successful implementation of user authentication, memory processing workflows, multimedia enhancement, AI-assisted narrative generation, secure data management, and responsive user interfaces across desktop and mobile environments.

The results indicate that integrating multiple AI technologies within a single platform can effectively transform fragmented personal information into meaningful and contextually rich memory narratives. ReMIND demonstrates the practical feasibility of AI-assisted memory reconstruction as a cognitive support tool and establishes a strong foundation for future research, large-scale deployment, and potential clinical adaptation in memory rehabilitation and assisted-living environments.

Dedication

This project is dedicated to all individuals and families affected by memory loss disorders such as Alzheimer's disease, dementia, and age-related cognitive decline. Their courage, resilience, and determination in facing the challenges of memory loss served as the inspiration behind ReMIND. We recognize that memories are far more than recollections of past events; they are an essential part of personal identity, relationships, and the experiences that shape our lives.

We would like to make a special dedication to a family member whose struggle with Alzheimer's disease inspired the creation of this project. Witnessing firsthand the emotional impact of memory loss and its effect on both the individual and their loved ones highlighted the need for innovative solutions that can help preserve meaningful experiences and strengthen human connections. This personal experience motivated us to explore how technology and artificial intelligence could be used to support individuals in reconnecting with their past.

This work is also dedicated to caregivers and loved ones who stand beside individuals experiencing cognitive decline. Their patience, compassion, and unwavering support play a vital role in preserving dignity, comfort, and a sense of belonging for those affected by memory-related disorders. Their dedication reminds us that memory is not only a personal experience but also a shared bond that connects people across generations.

Finally, we dedicate this project to the future of assistive healthcare technology. We hope that ReMIND serves as a small step toward the development of intelligent systems that improve quality of life, support emotional well-being, and help preserve the stories, experiences, and memories that define who we are. May this work inspire further research and innovation aimed at creating meaningful technological solutions for individuals facing the challenges of memory loss.

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Furthermore, we would like to acknowledge and appreciate the collaborative efforts of our team members. The successful completion of ReMIND is the result of teamwork, mutual trust, dedication, and countless hours of cooperation. Each member contributed their knowledge, skills, and commitment toward a shared vision, making this project a rewarding and enriching experience for all of us.

We also extend our sincere thanks to the University of Central Punjab for providing an excellent academic environment, valuable learning opportunities, technical resources, and the platform necessary to undertake and complete this project. The knowledge and experiences gained during our time at the university have been instrumental in shaping our academic and professional development.

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Revision History

Name	Date	Reason For Changes	Version

Chapter 1. Introduction

ReMIND is an AI-assisted web application designed to help individuals experiencing memory loss, including patients affected by Alzheimer's disease, dementia, and age-related cognitive decline. The system reconstructs fragmented memories using multimodal inputs such as text, images, and voice recordings. By combining artificial intelligence with an elderly-friendly user experience, ReMIND generates meaningful memory narratives that help patients re-live past experiences and strengthen emotional connections with caregivers and family members. The project is motivated by the emotional, cognitive, and social challenges associated with memory loss and aims to provide accessible technological support through intelligent memory reconstruction.

1.1 Product (Problem Statement)

ReMIND is an intelligent web-based memory reconstruction platform that assists individuals suffering from memory-related disorders by transforming fragmented information into meaningful memories. The system accepts multimodal inputs including text descriptions, photographs, and voice recordings, which are processed through integrated artificial intelligence services. Using speech recognition, image enhancement, contextual understanding, and narrative generation techniques, ReMIND reconstructs memory artifacts that users can revisit through an interactive chat-based interface. The platform also supports reminders, memory timelines, caregiver collaboration, secure authentication, and cloud-based data management. Through its combination of AI-powered reconstruction and human-centered design, ReMIND helps bridge emotional and cognitive gaps while improving memory recall and quality of life.

1.2 Background

The project lies at the intersection of artificial intelligence, cognitive rehabilitation, human-computer interaction, and healthcare technology. It applies modern AI techniques including natural language processing, speech processing, image enhancement, and generative AI to support individuals affected by memory decline. By integrating multiple AI services into a unified platform, ReMIND aims to improve memory recall, emotional well-being, and caregiver involvement while demonstrating the potential of intelligent systems in cognitive support and digital healthcare solutions.

1.3 Objective(s)/Aim(s)/Target(s)

The primary objective of ReMIND is to design, develop, and deploy an AI-assisted memory reconstruction platform that enables individuals experiencing memory loss to reconstruct meaningful memories from fragmented multimedia information. The project combines artificial intelligence, accessibility, and empathetic design principles to provide cognitive support for patients while enabling caregivers to participate in the memory reconstruction process.

The specific objectives are as follows:

Memory Reconstruction: Transform incomplete data such as text notes, photos, and voice recordings into coherent and contextually rich memory narratives.

Elderly-Friendly Interface: Design a simple, accessible interface optimized for older users and non-technical caregivers.

Secure Data Handling: Implement encryption, authentication, and cloud storage to protect sensitive personal data.

Proof of Concept Demonstration: Deliver a functional prototype using existing AI APIs and web technologies to validate the concept.

Overall, ReMIND seeks to enhance emotional well-being, cognitive support, and human connection through accessible AI innovation.

1.4 Scope

The scope of ReMIND includes the design, implementation, testing, and deployment of a web-based memory reconstruction system that processes text, image, and audio inputs to generate meaningful memory narratives. The system incorporates AI-powered memory reconstruction, multimedia enhancement, contextual analysis, reminder management, caregiver collaboration, user authentication, cloud storage integration, and timeline generation. The platform is designed with accessibility considerations for elderly users and supports secure handling of personal data. ReMIND is developed as a proof-of-concept solution intended to demonstrate the practical feasibility of AI-assisted memory reconstruction. Medical diagnosis, hospital information system integration, and real-time biometric monitoring remain outside the project scope.

1.5 Business Goals

As a successfully implemented proof-of-concept platform, ReMIND demonstrates the practical feasibility of AI-driven memory reconstruction for individuals experiencing memory loss. The project validates how modern AI technologies can be integrated to provide emotional and cognitive support through memory rebuilding, contextual recall, and caregiver collaboration. ReMIND establishes a foundation for future research, commercialization, and clinical adaptation within healthcare and assisted-living environments. By combining accessibility, empathy, and artificial intelligence, the project highlights opportunities for innovative digital health solutions that improve patient well-being and strengthen human connections through preserved memories.

1.6 Challenges

The development of ReMIND involved several technical, operational, and research-related challenges. One of the major challenges was integrating multiple AI services for text processing, voice processing, and image enhancement into a single unified platform. Since each AI service follows different APIs, response formats, and processing requirements, significant effort was required to ensure smooth communication and orchestration between system components. Another challenge was handling multimedia data efficiently while maintaining acceptable processing times and system responsiveness. Image enhancement and voice processing tasks were computationally intensive and required optimization to balance performance and quality. Data privacy and security were also critical concerns due to the sensitive nature of personal memories and user information, requiring secure authentication, cloud storage, and protected communication channels. Additionally, deploying and maintaining the complete system on a VPS environment while ensuring compatibility between frontend, backend, database, and AI modules presented several configuration and infrastructure challenges. Despite these difficulties, all core functionalities were successfully implemented and integrated into the final prototype.

1.7 Learning Outcomes

The development of ReMIND provided valuable practical experience in software engineering, artificial intelligence, and full-stack application development. The project enhanced our understanding of system architecture design, API integration, database management, cloud deployment, and secure application development. We gained hands-on experience working with modern technologies such as React.js, FastAPI, PostgreSQL, Cloudinary, and VPS hosting environments. The project also strengthened our knowledge of AI-powered systems through the integration of language models, speech recognition, voice synthesis, image enhancement, and retrieval-based processing techniques. Furthermore, we developed skills in project planning, teamwork, version control, testing, debugging, documentation, and deployment. Beyond technical knowledge, the project improved our problem-solving abilities, collaboration skills, and understanding of how technology can be applied to address real-world healthcare and social challenges. Overall, ReMIND provided a comprehensive learning experience that bridged academic concepts with practical industry-level implementation.

1.8 Nature of End Product

ReMIND is a fully functional web-based proof-of-concept platform designed to assist individuals experiencing memory loss through AI-assisted memory reconstruction. The final product accepts multimodal inputs including text, images, and voice recordings, processes them through specialized AI modules, and generates meaningful memory narratives presented through an interactive chat-based interface. The system includes user authentication, Google OAuth integration, multimedia processing, memory storage, timeline generation, administrative controls, and secure cloud-based media management. The platform is accessible through modern web browsers on both desktop and mobile devices and is deployed on a VPS hosting environment. Although developed as a proof-of-concept, the system demonstrates the practical feasibility of combining artificial intelligence, multimedia processing, and human-centered design to support memory recall and emotional well-being. Its modular architecture allows future scalability, feature expansion, and potential adaptation for larger healthcare and assisted-living applications.

1.9 Related Work/ Literature Survey/ Literature Review

Research in cognitive support systems has demonstrated the potential of artificial intelligence in assisting individuals with memory-related disorders. Existing solutions primarily focus on reminder systems, digital diaries, note-taking applications, and caregiver-assisted memory aids. While these tools help users manage daily activities, they generally depend on manually recorded information and provide limited assistance when memory details are incomplete or unavailable. Recent advancements in natural language processing, computer vision, speech recognition, and generative AI have created opportunities for more intelligent memory support systems. Studies involving AI-powered conversational assistants, image understanding models, and contextual reasoning systems have shown promising results in enhancing user interaction and information retrieval. ReMIND builds upon these developments by combining multimodal inputs, AI-assisted contextual understanding, image enhancement, speech processing, and narrative generation within a single platform. Unlike traditional reminder-based systems, ReMIND focuses on reconstructing meaningful memories from fragmented information, thereby contributing to a more holistic and emotionally supportive approach to cognitive assistance and digital memory preservation.

1.10 Document Conventions

The Document Conventions being used throughout the document are following:

- Font Style being used is Times New Roman.
- Main Headings are of Size 16.
- Sub-Headings are of Size 14.
- Content/Paragraph and Tertiary headings are of Size 12.
- Auto line spacing is used.

Chapter 2. Overall Description

2.1 Product Features

ReMIND is an AI-assisted memory reconstruction platform designed to help individuals experience memory loss by transforming fragmented information into meaningful memories. The system supports multimodal input through text, images, and voice recordings. Users can upload personal memories, photographs, and audio clips, which are processed using specialized AI modules for contextual understanding, image enhancement, speech recognition, and narrative generation.

Key features include secure user authentication, Google OAuth integration, multimedia processing, AI-powered memory reconstruction, memory history management, cloud-based media storage, and an interactive chat interface. The platform also provides image enhancement, face restoration, speech-to-text conversion, voice synthesis, and contextual memory generation capabilities. Administrative features allow monitoring of users, system activity, audit logs, and platform analytics.

The system is accessible through modern web browsers and is designed with usability and accessibility considerations for elderly users. ReMIND serves as a proof-of-concept demonstrating how artificial intelligence can be applied to support cognitive assistance, memory preservation, and emotional well-being through intelligent reconstruction of personal experiences.

2.2 Functional Description

ReMIND operates as a web-based client-server application that enables users to reconstruct memories from multimedia information. Users first authenticate through the system and then provide memory-related content in the form of text, images, or audio recordings. The uploaded data is validated and processed by the backend services before being forwarded to specialized AI modules for analysis and enhancement.

The text processing module performs contextual understanding and memory generation; the voice processing module converts speech into text and generates audio responses, while the image processing module enhances photographs and extracts visual context. The processed outputs are combined by the memory reconstruction engine to generate a coherent memory narrative.

The generated memories are stored in the database and can later be retrieved through the memory history interface. Administrative functions allow monitoring of system activity, user management, and audit logging. The overall workflow follows a secure, event-driven architecture where frontend, backend, database, cloud storage, and AI services collaborate to provide an integrated memory reconstruction experience.

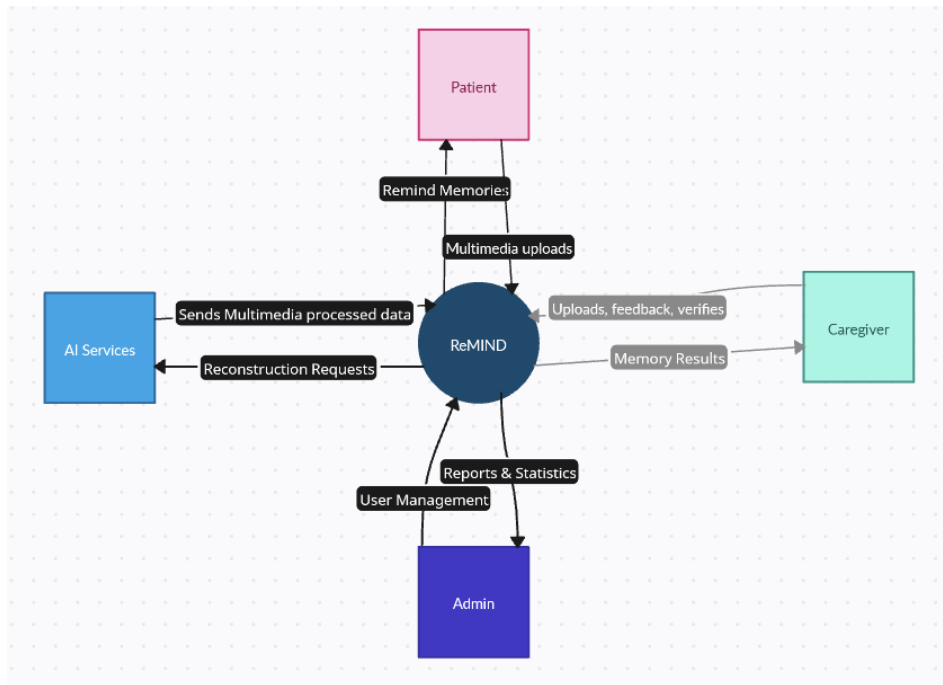


Fig 2.2: Context Diagram

2.3 User Classes and Characteristics

The ReMIND system supports two primary user classes: Users and Administrators.

Users represent the primary target audience of the system. These users may include individuals experiencing memory-related difficulties, elderly users, or family members assisting with memory reconstruction. Users interact with the platform by registering accounts, uploading text, images, and voice recordings, viewing reconstructed memories, and accessing memory history. Since many users may possess limited technical expertise, the interface is designed to be simple, accessible, and easy to navigate.

Administrators are responsible for managing and monitoring the platform. Their responsibilities include viewing registered users, monitoring system activity, accessing audit logs, reviewing platform analytics, and maintaining system integrity. Administrators require a higher level of access than standard users and interact with dedicated management interfaces.

Users are considered the highest-priority user class because the primary objective of the project is to support memory reconstruction and cognitive assistance. Administrative users are important for maintenance and monitoring but are secondary to the system's core purpose.

2.4 Design and Implementation Constraints

Several constraints influenced the design and implementation of ReMIND. Since the project was developed as a Final Year Project, proof-of-concept, development time, computational resources, and budget limitations affected technology selection and feature scope. The system relies on existing AI models and third-party APIs rather than developing custom machine learning models from scratch.

Technical constraints required the use of specific technologies including React.js for frontend development, FastAPI for backend services, PostgreSQL for database management, Cloudinary for media storage, and Hostinger VPS for deployment. Integration with external AI services such as Groq Llama 3, ElevenLabs, Google Text-to-Speech, GFPGAN, Real-ESRGAN, and BLIP introduced dependencies on API availability and internet connectivity.

Security requirements mandated authentication, authorization, protected routes, and secure communication through HTTPS and RESTful APIs. Browser compatibility requirements also influence interface design to ensure accessibility across desktop and mobile devices. Additionally, some planned features such as advanced reminder scheduling and full caregiver role integration were limited by project scope and time constraints.

2.5 Assumptions and Dependencies

The development of ReMIND is based on several assumptions. It is assumed that users have access to a stable internet connection, a modern web browser, and devices capable of uploading images and audio recordings. It is also assumed that users can provide sufficient memory-related information to allow meaningful reconstruction by the AI modules.

The system depends heavily on third-party technologies and services. These include Groq Llama 3 for text processing and memory reconstruction, ElevenLabs for speech recognition, Google Text-to-Speech for voice synthesis, GFPGAN and Real-ESRGAN for image enhancement, BLIP for image captioning, Cloudinary for multimedia storage, PostgreSQL for data persistence, and Hostinger VPS for hosting and deployment. Any service interruption, API modification, or policy change affecting these technologies may impact system functionality.

The project further assumes continued compatibility between integrated software libraries, frameworks, and operating environments. Since ReMIND is implemented as a proof-of-concept, future scalability, healthcare integration, multilingual support, and advanced notification services remain dependent on future development efforts and resource availability.

Chapter 3. System Requirements

3.1 Functional Requirements

3.1.1 Upload Multimedia Input

Identifier	Multimedia Uploading	
Purpose	To allow users to upload photos, audio, or text that will be the input for the memory OR to enter text.	
Priority	High	
Pre-conditions	Users must be logged into the system. The device must have a stable internet connection.	
Post-conditions	Uploaded multimedia files are successfully stored in Firebase/Cloudinary, and metadata is recorded in PostgreSQL.	
Typical Course of Action		
S#	Actor Action	System Response
1	User logs into the ReMIND web application.	System verifies credentials and loads the dashboard.
2	User selects the Upload Memory option.	System displays upload options for photo, audio, or text.
3	The user confirms the upload.	System stores the file in cloud storage and logs its metadata.
4	System notifies users of successful uploads.	Confirmation message displayed and file becomes available for memory reconstruction.
Alternate Course of Action		
S#	Actor Action	System Response
1	User uploads an invalid or unsupported file.	System displays an error message and prompts re-upload.
2	Internet connection is lost.	System pauses upload and provides retry options once connection is restored.

Fig 3.1.1: UC – Table 1

3.1.2 Generate Reconstructed Memory

Identifier	Memory Reconstruction	
Purpose	To generate a meaningful event summary by processing multimedia inputs using AI services.	
Priority	High	
Pre-conditions	Users have uploaded multimedia content. AI APIs are accessible and functional.	
Post-conditions	Reconstructed memory is stored and displayed to the user in the timeline.	
Typical Course of Action		
S#	Actor Action	System Response
1	User selects uploaded files for reconstruction.	The system retrieves the selected files from storage.
2	User clicks Generate Memory .	The system sends files to integrated APIs (Whisper, Vision, etc.) for analysis.
3	The user waits for the result.	The system processes text, image, and audio data.
4	User reviews the generated memory.	The system displays reconstructed narrative, enhanced image, and optional audio playback.
5	The user saves memory.	The system stores the final output in the database and updates the user's timeline.
Alternate Course of Action		
S#	Actor Action	System Response
1	AI services fail or timeout.	The system shows a retry option and logs the error.
2	Uploaded files are incomplete or corrupted.	The system will tell the user to verify the files.

Fig 3.1.2: UC – Table 2

3.1.3 Manage Users and System Data

Identifier	System Data Management	
Purpose	To allow the administrator to manage user accounts, monitor system activity, and configure AI services within ReMIND app.	
Priority	High	
Pre-conditions	Admin must log in with valid credentials and have access privileges.	
Post-conditions	All user or system configuration changes are updated in the database.	
Typical Course of Action		
S#	Actor Action	System Response
1	Admin logs into the ReMIND	The system verifies credentials and loads the admin dashboard.
2	Admin selects a user account to edit, disable, or remove.	The system validates the action and updates the user's status accordingly.
3	Admin reviews and updates API usage statistics or system logs.	The system fetches the requested logs and updates them.
Alternate Course of Action		
S#	Actor Action	System Response
1	Admin attempts to delete a protected or active user account.	The system denies the action and displays a warning message.
2	Admin session expires during activity.	The system logs out the admin and asks for authentication again.

Fig 3.1.3: UC – Table 3

3.1.4 Requirements Analysis and Modeling

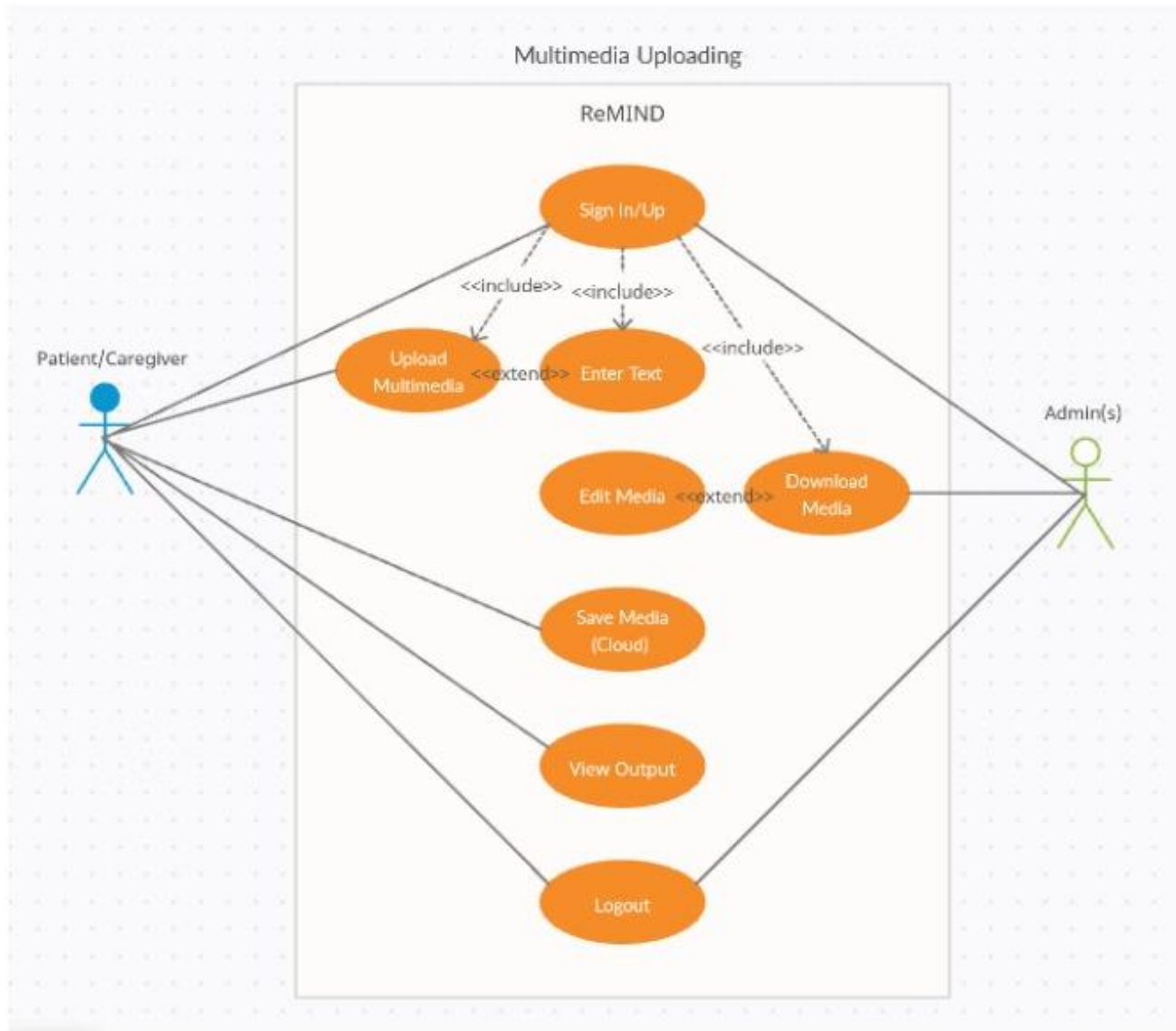


Fig 3.1.4.1: Use Case 1

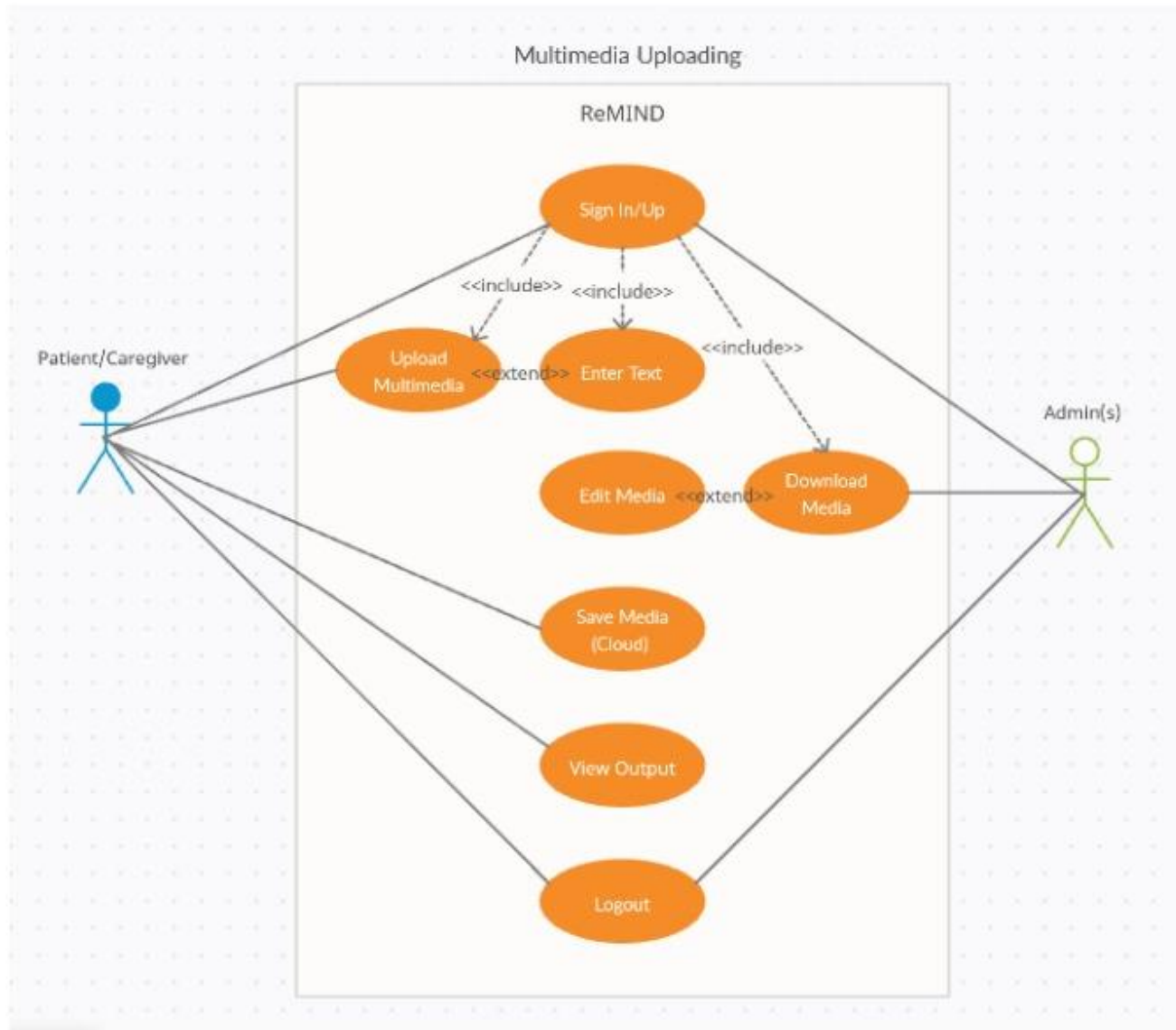


Fig 3.1.4.2: Use Case 2

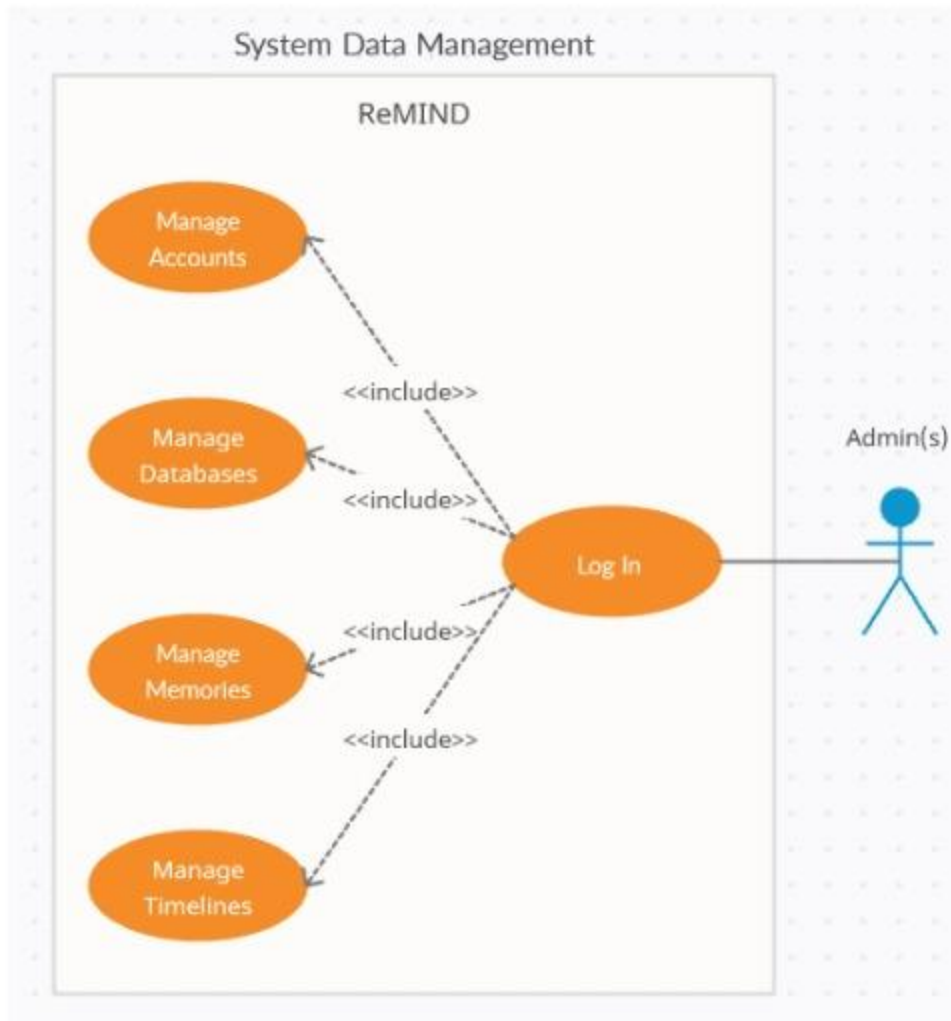


Fig 3.1.4.3: Use Case 3

3.2 Nonfunctional Requirements

3.2.1 Performance Requirements

The ReMIND system shall provide responsive and efficient memory reconstruction services for users accessing the platform through modern web browsers. User authentication requests should be processed within 3 seconds under normal operating conditions. Text-based memory reconstruction responses should be generated within 10 seconds, while image and audio processing tasks should be completed within 30 seconds depending on file size and external AI service response times. The system shall support concurrent access by at least 50 users without significant degradation in performance. Database queries for memory retrieval, and user information shall return results within 2 seconds. Media uploads shall support files up to 25 MB and provide upload status feedback to users. The platform shall maintain availability of at least 95% during the deployment period. Performance monitoring logs shall be maintained to identify bottlenecks and ensure efficient operation of AI processing, database transactions, and API communications.

3.2.2 Safety Requirements

Since ReMIND is a cognitive support application and not a medical diagnostic system, the platform shall not provide medical diagnoses, treatment recommendations, or healthcare decisions. The system shall clearly communicate that generated memory narratives are AI-assisted reconstructions and may require user verification. Users shall retain control over all uploaded and generated content. The system shall prevent accidental deletion of memory records through confirmation prompts before destructive actions are executed. Uploaded multimedia files shall be validated before processing to prevent corruption and system failures. Error handling mechanisms shall notify users when AI services are unavailable, or processing fails. The platform shall maintain backup procedures for stored memory data to reduce the risk of data loss. Any generated content that could potentially misrepresent memories shall be presented as reconstructed information rather than verified facts.

3.2.3 Security Requirements

The ReMIND system shall implement secure user authentication using JWT-based authentication and Google OAuth integration. User passwords shall be securely hashed before storage and shall never be stored in plain text. All communication between the client and server shall occur over HTTPS connections. Role-based access control shall ensure that only authorized users can access protected resources and administrative functions. User sessions shall automatically expire after a defined period of inactivity. Sensitive personal information, uploaded media, and generated memories shall be stored securely within PostgreSQL and Cloudinary services. The system shall validate all user inputs to prevent common attacks such as SQL injection, cross-site scripting, and unauthorized file uploads. Audit logs shall be maintained to track significant user and administrative activities. The platform shall comply with general privacy and data protection principles by restricting unauthorized access to user data.

3.2.4 Additional Software Quality Attributes

Usability: The system shall provide an elderly-friendly and intuitive interface with readable typography, simplified navigation, and responsive design for desktop and mobile devices.

Reliability: The platform shall successfully complete at least 95% of valid processing requests under normal operating conditions.

Maintainability: The modular architecture shall separate frontend, backend, AI services, and database layers to simplify future maintenance and enhancements.

Scalability: The architecture shall support future expansion through independent AI modules, cloud storage services, and database scaling.

Portability: The application shall operate on major web browsers including Chrome, Firefox, Edge, and Safari without requiring additional software installation.

Availability: The deployed system shall remain accessible through internet-based services and recover gracefully from service interruptions.

Interoperability: The platform shall integrate with external AI services, cloud storage providers, and RESTful APIs through standardized communication protocols.

Testability: System modules shall support independent testing and validation through API endpoints, logging mechanisms, and test cases.

3.3 Other Requirements

The system shall utilize PostgreSQL as the primary relational database for storing user profiles, memory records, metadata, audit logs, and application data. Cloudinary shall be used for cloud-based storage and delivery of multimedia content. The platform shall communicate with external AI services through secure RESTful APIs. ReMIND shall support modern web browsers and require only internet connection, microphone access, and multimedia upload capabilities from client devices. The system shall be developed using React.js for the frontend and FastAPI for backend services. All source codes shall be maintained using Git-based version control. The platform shall support future integration with healthcare systems, multilingual processing services, mobile applications, and advanced notification modules. Reusable open-source technologies including Groq Llama 3, ElevenLabs, GFPGAN, Real-ESRGAN, BLIP, SQL Alchemy, and Alembic shall be incorporated where appropriate to reduce development effort and improve maintainability.

Chapter 4. Technical Architecture

ReMIND is a specially designed web application that makes use of both **open-source AI models** and **Commercial Off-The-Shelf (COTS) AI services**. The system incorporates specialized technologies for natural language processing, speech recognition, voice synthesis, image enhancement, and contextual understanding rather than creating machine learning models from the ground up. This strategy allowed for quick development while guaranteeing dependable performance and excellent results within the technical and academic limitations of the project. ReMIND supports online, nearly real-time processing through a **layered client-server architecture**. User authentication, multimedia uploads, memory reconstruction requests, timeline creation, reminder management, and caregiver interactions are among the event-driven transactions that the system manages. The system keeps track of audit logs, feedback records, and evaluation metrics for administrative and monitoring purposes in addition to transactional processing.

Key Application Components and Duties:

The **Frontend Layer** uses **React** to create a browser-based user interface. Users can access timelines, manage reminders, upload multimedia content, engage with reconstructed memories, and communicate via a chat-based interface. With usability considerations for senior users and caregivers, the interface is responsive and **optimized for desktop and mobile devices**.

FastAPI is used to implement the **Backend Layer**, which acts as the platform's central control unit. The AI Processing Layer consists of specialized Text, Voice, and Image Processing modules. Memory reconstruction, retrieval-augmented generation, contextual understanding, and natural language processing are all performed by the Text Processing Module. Speech-to-text conversion, emotion and tone analysis, and voice synthesis are all performed by the Voice Processing Module. Image enhancement, face restoration, image caption generation, and visual context extraction are all performed by the Image Processing Module.

Both structured and unstructured data produced throughout the system are managed by the Data Layer. The **main database** used to store user accounts, memory artifacts, timelines, audit logs, feedback, reminders, and metadata is **PostgreSQL**.

The finished architecture effectively combines database management, cloud storage, AI modules, frontend services, and backend processing into a single ecosystem that can reconstruct meaningful memories from multimodal inputs. Beyond the proof-of-concept phase, the modular design facilitates future expansion, scalability, and maintainability.

4.1 Application and Data Architecture

The ReMIND system follows a layered, modular, and service-oriented architecture designed to support intelligent memory reconstruction through the integration of multiple processing components. The architecture consists of four primary layers: the Frontend Layer, Backend Layer, AI Processing Layer, and Data Layer. The Frontend Layer provides a browser-based user interface through which users interact with the system, submit memory-related content, view reconstructed memories, and manage their profiles. The Backend Layer serves as the central coordination hub, handling authentication, request processing, business logic, API communication, and orchestration of system services.

The AI Processing Layer contains specialized modules responsible for text processing, voice processing, and image processing. These modules collaborate to analyze user-provided multimedia content, extract contextual information, enhance media quality, and generate meaningful memory narratives. The Data Layer manages both structured and unstructured information, including user profiles, memory artifacts, audit logs, metadata, and multimedia assets. PostgreSQL is used for persistent data storage, while Cloudinary provides secure cloud-based media storage and delivery.

The architecture emphasizes modularity, scalability, maintainability, and interoperability by separating responsibilities among independent components and enabling communication through standardized APIs. This design allows future enhancements and integration of additional AI services without major architectural modifications.

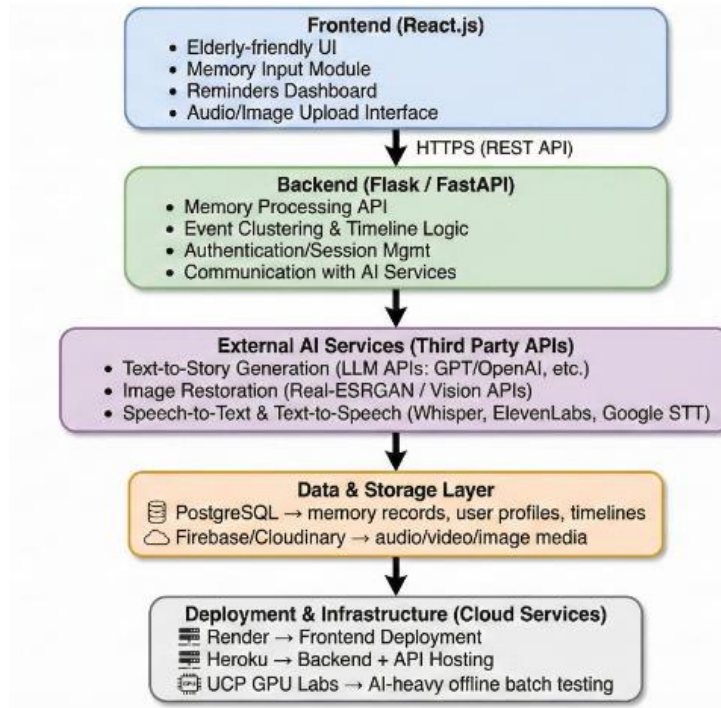


Fig 4.1: Technical Architecture

4.2 Component Interactions and Collaborations

The ReMIND system operates through coordinated interactions between users, application services, databases, cloud storage, and artificial intelligence modules. User requests originate from the frontend interface and are transmitted to the backend services for authentication, validation, and processing. The backend then communicates with the appropriate AI modules based on the type of multimedia content provided. Text inputs are forwarded to the text processing service for contextual understanding and memory generation, audio inputs are processed through speech recognition and voice synthesis services, while image inputs are analyzed through enhancement and caption generation modules.

After processing is completed, the generated outputs are aggregated by the memory reconstruction engine to create a coherent memory narrative. The reconstructed memory and associated metadata are then stored within the database and media storage services. Users can subsequently retrieve, review, and interact with previously generated memories through the application interface. Administrative components simultaneously monitor system activities, maintain audit logs, and support management operations.

The interaction model follows a secure and event-driven workflow where each component performs a specialized function while collaborating through clearly defined interfaces. This approach improves system reliability, simplifies maintenance, and enables efficient coordination between internal services and external AI technologies.

Component Diagram:

This Component Diagram illustrates the physical architecture of the ReMIND system, showing how the software components interact. The core communication hub is the API Gateway, which routes requests between different modules. The Frontend (React.js) provides user interfaces like the Memory Input UI and Reconstructed Memories Viewer. The Backend FastAPI handles User Management and Event Clustering. The system relies on external AI Services for tasks like Story Generation and Speech-to-Text. Data is persistently stored in a PostgreSQL Database and Cloudinary storage. The Notification Module handles reminders, and the entire system is deployed using VPS.

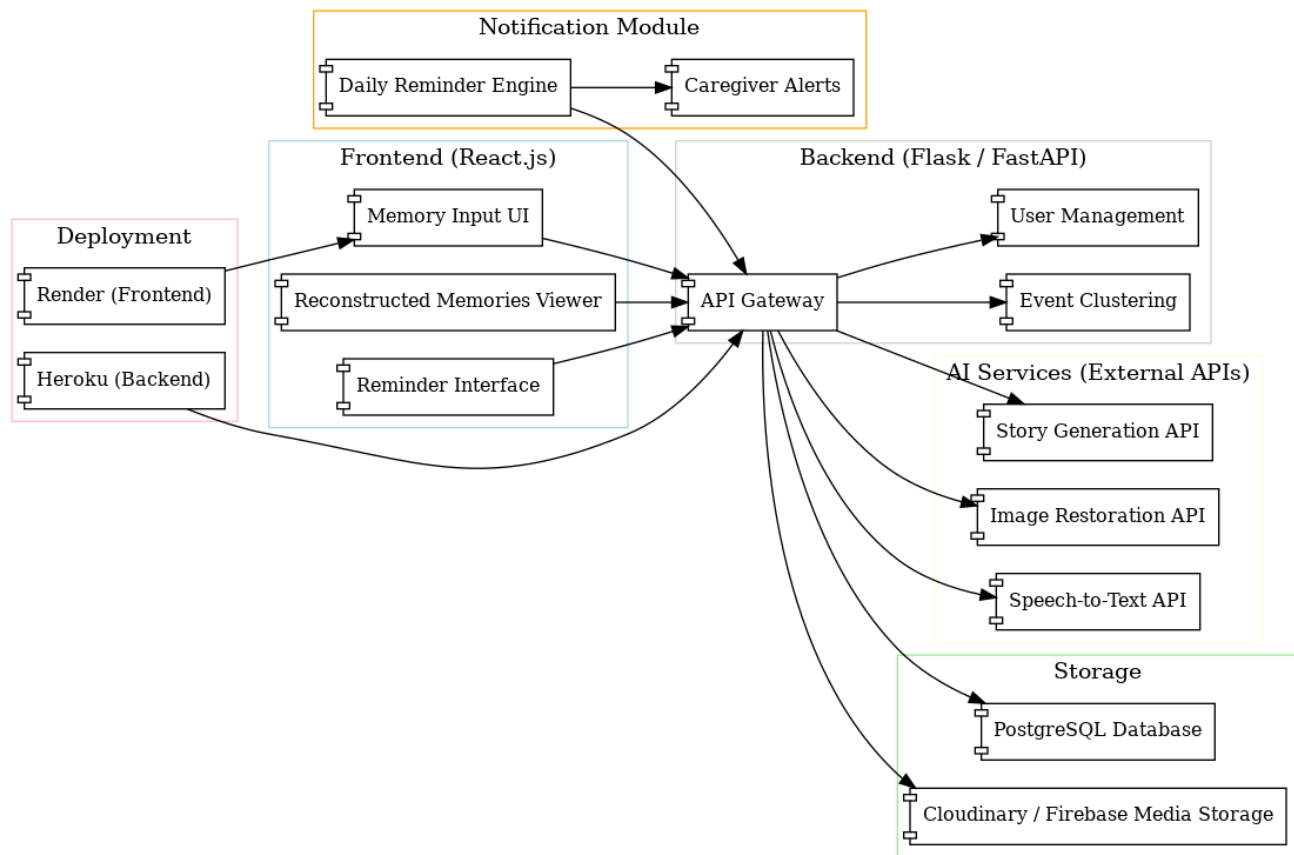


Fig 4.2.1: Component Diagram

Collaboration Diagram:

This Collaboration Diagram outlines the functional flow of the ReMIND system. It begins with the User initiating uploadMemory() (1) through the Frontend, which triggers sendJSON() (2) to the API Gateway. After the Auth Service performs authenticatesUser() (3), the MPC manages the Memory Reconstruction phase by identifying the input type (5) for the Text, Image, or Audio AI Engines. Once processed, the reconstructedOutput() (6) is sent to the Evaluation Engine to calculateAccuracy() (7) before being committed to the Database. The cycle completes as the system storeLogs() (8) and transmits a requestJSON() (9) back to the Frontend for final displayMemory() (10) via the Display Output module.

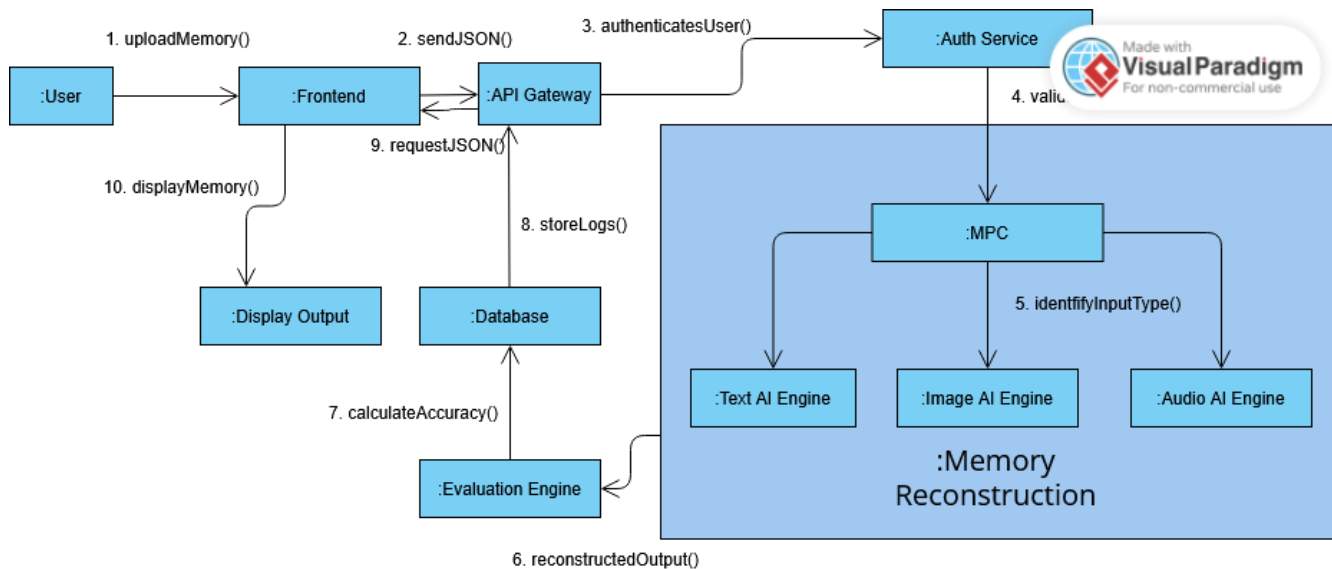


Fig 4.2.2: Collaboration Diagram

ERD Diagram:

This Entity-Relationship Diagram (ERD) visualizes the structure of a database for ReMIND. It models key entities such as the adaptable User (specialized as Patient, Caregiver, and Admin) and the relationships that connect them. Patients, associated with Conditions, receive care and are managed by Caregivers. The operational core includes Chat Sessions for communication, composed of Messages and Media Files, with support for Reminders (which issue Notifications), and the capture of user Feedback. All critical activities are logged in an Audit Log, and the system includes functionality for processing media using AI Processing Tasks.

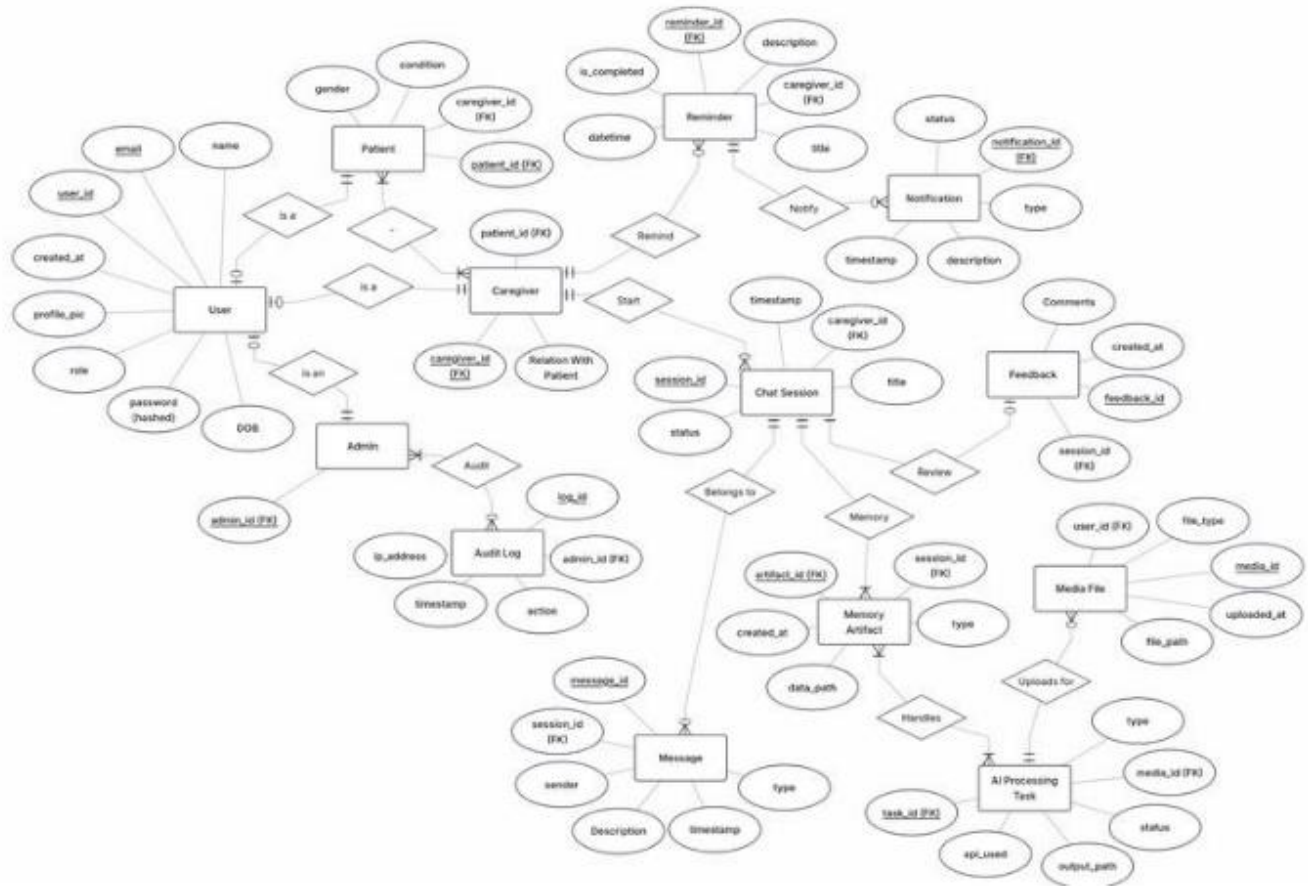


Fig 4.2.3: ERD Diagram

UML Class Diagram:

This UML Class Diagram models the structure of ReMIND. It organizes functionality into modules: Profiles, Audit Log, Caregiver & Patient, Chat, and AI & Media. The Profiles module defines User, specialized in Caregiver Profile, Patient Profile, and Admin Profile. The Caregiver & Patient Module manages Reminders and Notifications. The Chat Module handles Chat Sessions, Messages, and Feedback. Finally, the AI & Media Module defines Media, Memory Artifact, and AI Processing Task for handling media content.

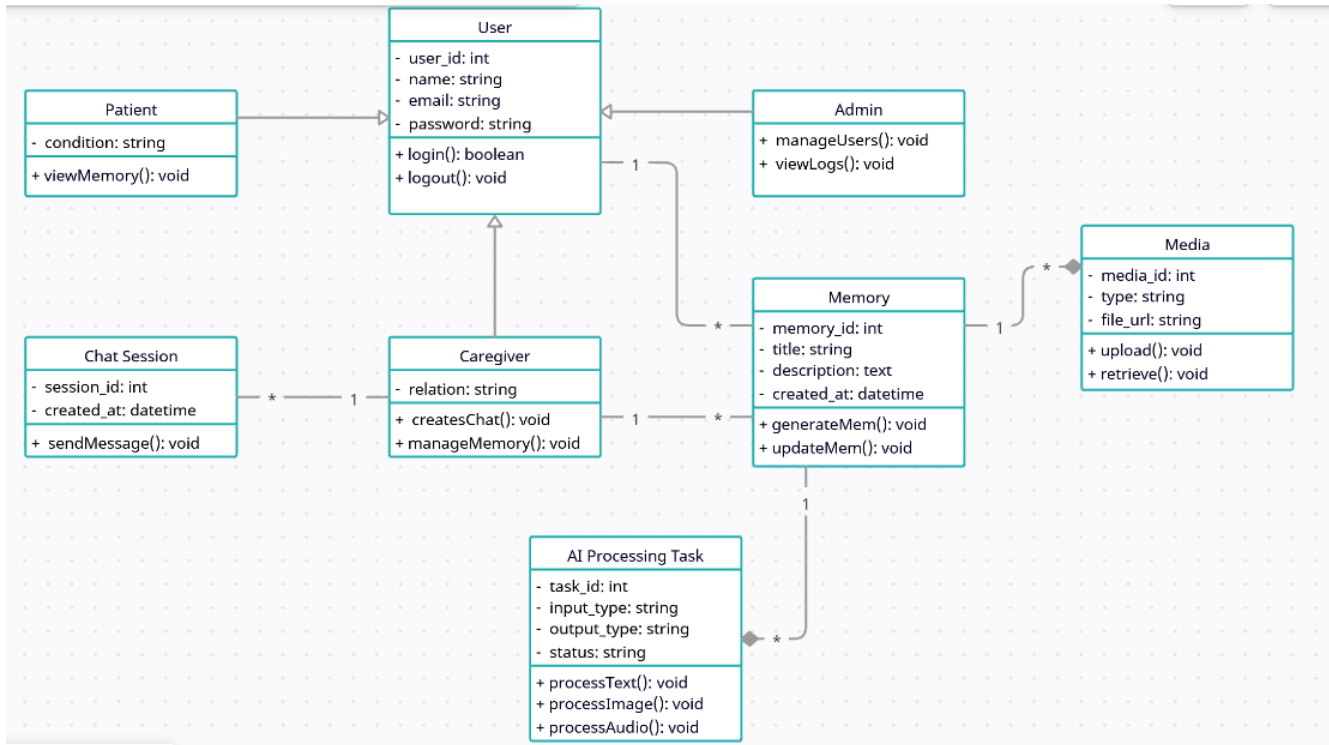


Fig 4.2.4: UML Class Diagram

Relational Schema:

This relational schema, based on the ReMIND-DATABASE ERD. It features a central **User** table, with specialized child tables: **AdminProfile**, **CaregiverProfile**, and **PatientProfile**, linked by foreign keys (FK) on the respective ID fields. The core interactions occur in the **ChatSession** table, which references **CaregiverProfile** and **PatientProfile**. Communication data is stored in **Message** and **Media** tables. Support functionality includes **Reminder** and **Notification** tables. All actions are tracked in the **Auditlog** table, while **AIProcessingTask** and **MemoryArtifact** handle media processing and retrieval.

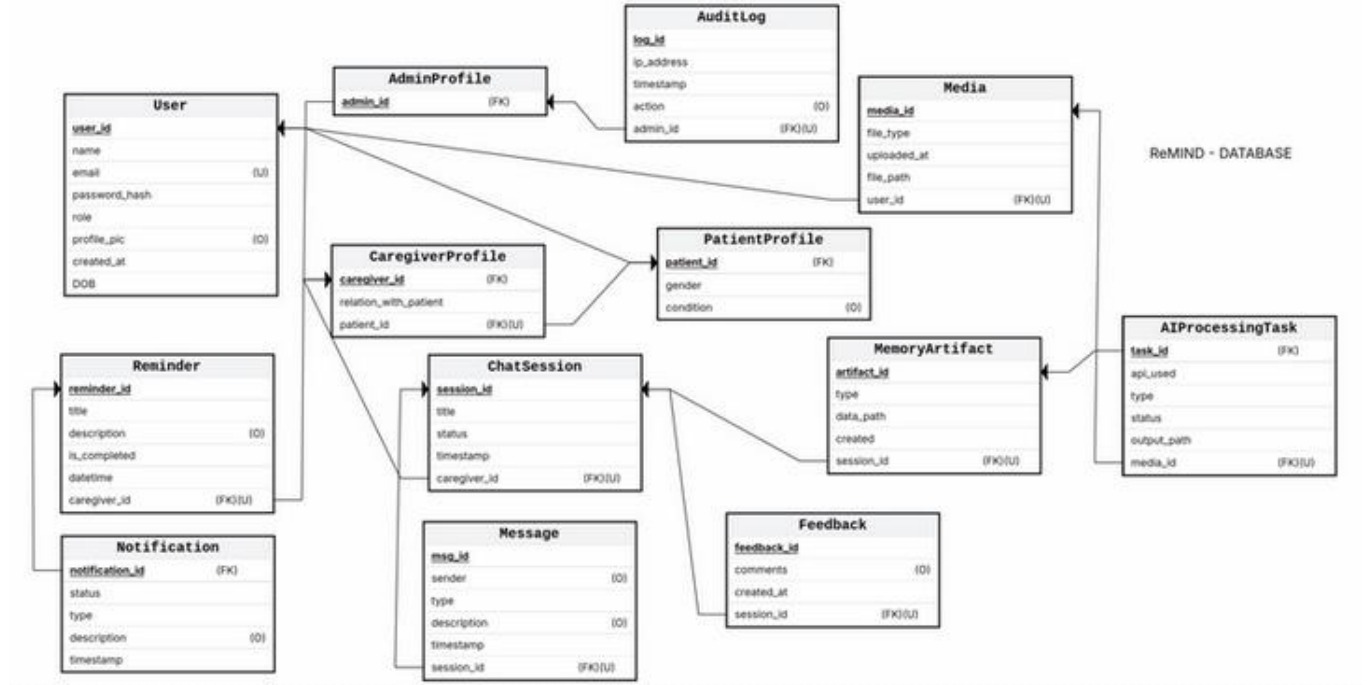


Fig 4.2.5: Relational Schema

Decision Table:

Rule	Input	Alternative Conditions	Action	Output
R1	User provides valid login credentials	Invalid credentials	Authenticate user, create session	Access granted / error message displayed
R2	Patient uploads multimedia input	Unsupported format or missing metadata	Accept input, store temporarily, trigger preprocessing	Acknowledgment / error message
R3	Uploaded file validated successfully	File corrupted or exceeds size limit	Proceed to AI preprocessing and enhancement	Process initiated / warning prompt
R4	Image input detected	No image detected	Trigger Real-ESRGAN enhancement and tagging	Enhanced image + metadata stored
R5	Audio input detected	No audio provided	Send to Whisper API for transcription	Transcript text returned
R6	Text or transcript available	Missing or empty text input	Send to GPT-4 for contextual understanding	Structured text/context response
R7	All media processed successfully	Any AI module fails	Aggregate data and reconstruct event	Reconstructed memory / retry message

ReMIND: Generative AI Based Memory Rebuilder

R8	Narrative generation successful	AI fails to generate or timeout	Store reconstructed memory in repository	Text + voice response / system retry
R9	Reminder set by caregiver	Reminder missing or invalid	Save reminder in database	Reminder confirmation / validation error
R10	Feedback submitted by patient/caregiver	Feedback skipped	Store feedback in log	Confirmation message / skip noted
R11	Admin monitors system logs	Logs not accessible	Display system status dashboard	Log summary / permission error

Fig 4.2.6: Decision Table

Event Table:

Event	Event Name	Trigger Type	Source	System Activity	Output
E1	User Login	External (User Action)	Patient / Caregiver / Admin	Authenticate credentials, create session	Dashboard or role-based access
E2	Media Upload	External (User Action)	Patient / Caregiver	Validate and store uploaded media	Temporary file storage + confirmation
E3	Data Preprocessing	Internal (System)	Backend	Cleans, formats, validates input data	Ready media for AI modules
E4	Image Enhancement	External (AI Service)	Real-ESRGAN	Enhances image and extracts context	Enhanced image file stored
E5	Audio Transcription	External (AI Service)	Whisper API	Converts audio to text, detects tone	Transcript returned
E6	Text Analysis	External (AI Service)	GPT-4	Interprets and structures text context	Semantic data created
E7	Event Reconstruction	Internal (AI Process)	AI Module	Combines multimodal data into cohesive memory	Structured narrative generated
E8	Voice Synthesis	External (AI Service)	ElevenLabs	Converts narrative text to natural audio	Voice file generated and stored
E9	Memory Storage	Internal (Database)	Backend → PostgreSQL	Saves reconstructed memory with metadata	Record created in memory repository
E10	Memory Retrieval	External (User Request)	Patient / Caregiver	Fetches requested memory or timeline	Displayed in frontend interface
E11	Reminder Creation	External (User Action)	Caregiver	Adds reminder entry in database	Confirmation sent to user
E12	Reminder Notification	Internal (Timed Event)	Notification Module	Sends scheduled alert to patient	Notification on UI
E13	Feedback Submission	External (User Action)	Patient / Caregiver	Logs ratings and suggestions	Stored in feedback log
E14	System Monitoring	Internal (Admin Action)	Administrator	Reviews logs, checks API integrations	Diagnostic report generated

Fig 4.2.7: Event Table

DFD Diagram Level 1:

The DFD Level 1 diagram illustrates the primary functional processes of the ReMIND system and the flow of data between users, databases, and AI services. The process begins with User Authentication (3.0), where Patients, Caregivers, and Administrators provide login or registration information. The system validates these credentials through the Accounts Database and grants authorized access to subsequent modules. Once authenticated, users can interact with the Data Processing module (1.0), which receives and validates uploaded multimedia and textual information. The processed data is stored in the Data Processing Database and forwarded to the Memory Reconstruction module (2.0). This module utilizes AI-powered services to analyze the processed information, reconstruct meaningful memories, and generate personalized results and feedback. The reconstructed memories are then stored in the Memory Artifacts Database for future retrieval, while analytical reports and system statistics are made available to administrators for monitoring and management purposes.

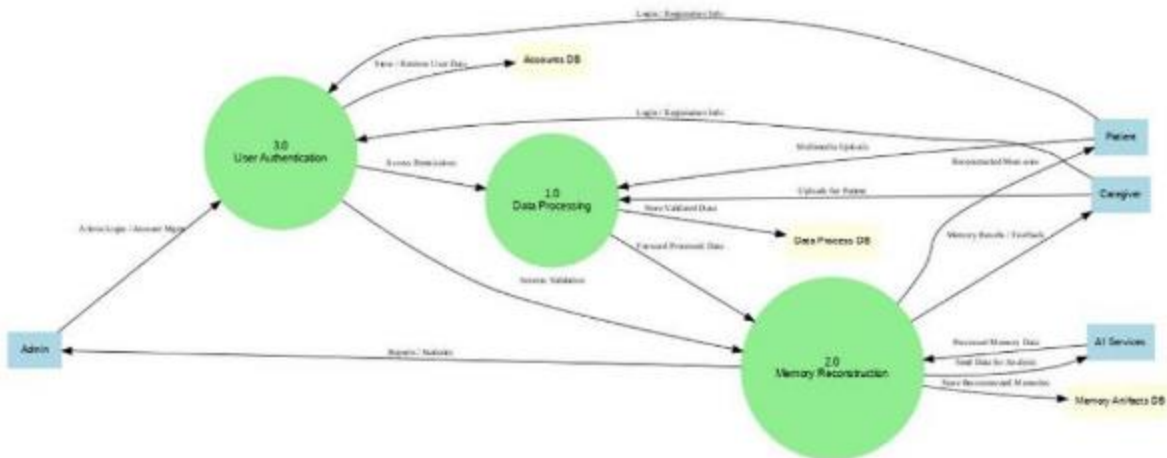


Fig 4.2.8: DFD Level 1

DFD Diagram Level 2:

The DFD Level 2 diagram provides a more detailed view of the internal operations of the ReMIND system by decomposing the Memory Reconstruction process into smaller functional components. The system begins with User Authentication (3.0), which validates user credentials and manages access permissions for Patients, Caregivers, and Administrators. Data Processing (1.0) receives multimedia and textual inputs from users, performs validation and preprocessing, and stores the processed information in the Data Processing Database. The processed data is then forwarded to the Memory Reconstruction module and its sub-processes for further analysis. Within Memory Reconstruction, the API Calls & Integration process (2.0.1) manages communication with external AI services. It sends relevant media and data for analysis, receives AI-generated outputs, and records all API interactions in the APIs Database for tracking and auditing purposes. The generated results are then utilized by the Memory Verification process (2.0.2), which evaluates and validates reconstructed memories using user-provided multimedia content. Verified memories are stored in the Memory Artifacts Database, while verification outcomes, feedback, and monitoring information are provided to the Administrator. This decomposition offers a clearer representation of how AI integration and verification mechanisms contribute to accurate memory reconstruction within the ReMIND system.

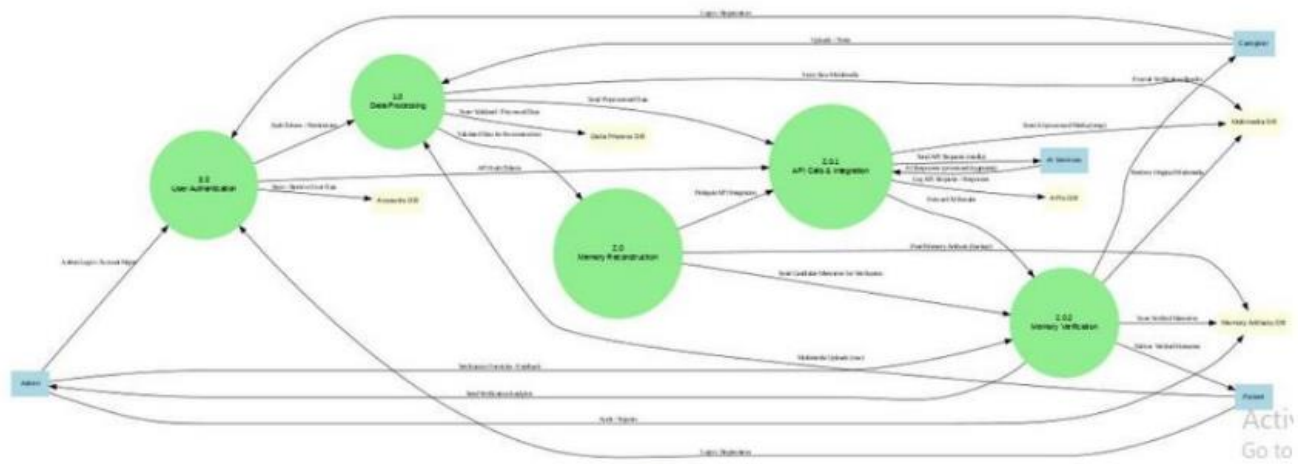


Fig 4.2.9: DFD Level 2

4.3 Design Reuse and Design Patterns

The development of ReMIND extensively utilizes software reuse principles to improve reliability, accelerate development, and ensure technical feasibility within the project's academic scope. Rather than developing every component from scratch, the system integrates well-established frameworks, libraries, cloud services, and pre-trained artificial intelligence models. This approach allowed the development team to focus on system integration, workflow orchestration, user experience, and memory reconstruction logic while leveraging proven technologies for supporting functionalities.

At the application layer, ReMIND reuses modern open-source frameworks including React.js for frontend development and FastAPI for backend services. These frameworks provide standardized routing, state management, API handling, middleware support, and security mechanisms that significantly reduce development complexity while improving maintainability and scalability.

At the intelligence layer, the system integrates pre-trained artificial intelligence models and external AI services rather than developing custom machine learning models. The Text Processing Module utilizes Groq-Llama 3 and Retrieval-Augmented Generation (RAG) techniques for memory reconstruction and contextual understanding. The Voice Processing Module leverages ElevenLabs STT for speech processing and Google TTS voice synthesis, while the Image Processing Module utilizes GFPGAN, Real-ESRGAN, and Blip-lightweight for image enhancement, face restoration, and image caption generation. Reusing these mature AI technologies eliminated the need for extensive model training and optimization while ensuring high-quality outputs.

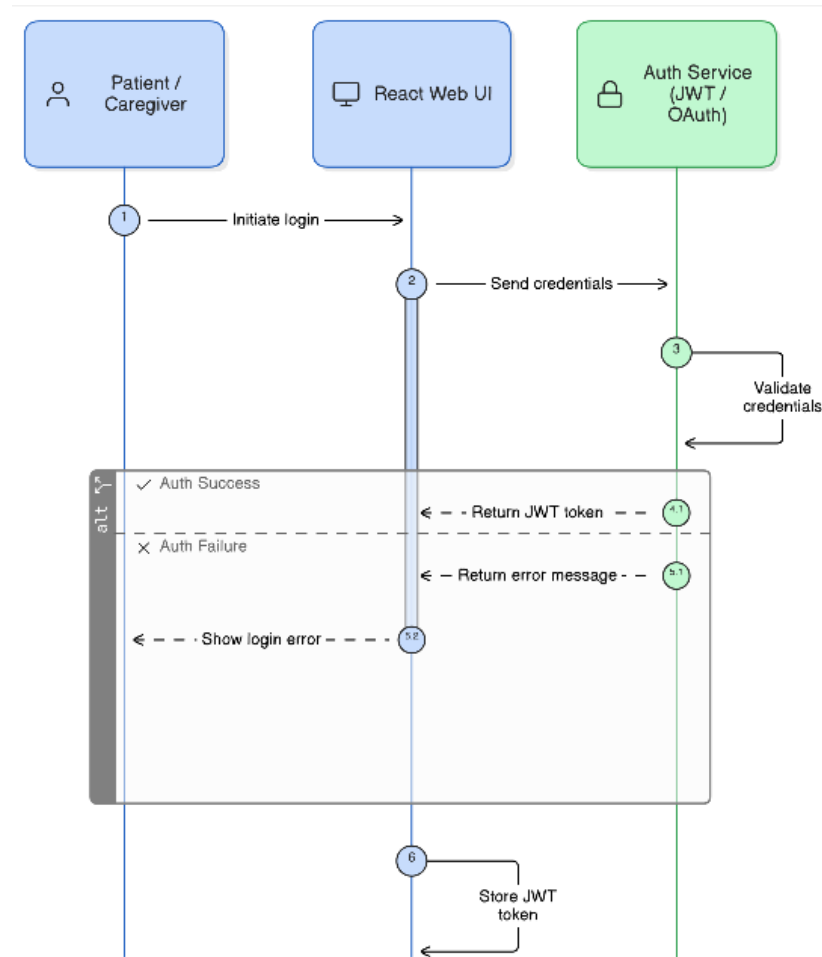
At the data layer, PostgreSQL is reused as a robust relational database management system for storing user information, memory artifacts, reminders, feedback, audit logs, and metadata. Cloudinary is utilized for secure cloud-based storage and delivery of multimedia assets. SQLAlchemy and Alembic are reused for database interaction and schema migration management.

Several software engineering and architectural design patterns were adopted throughout the implementation. The Layered Architecture Pattern separates the Presentation, Application, AI Processing, and Data layers to improve modularity and maintainability. The API Gateway and Service-Oriented approach is used within the backend to coordinate communication between frontend services, AI modules, databases, and cloud storage platforms. The Model-View-Controller (MVC) concept

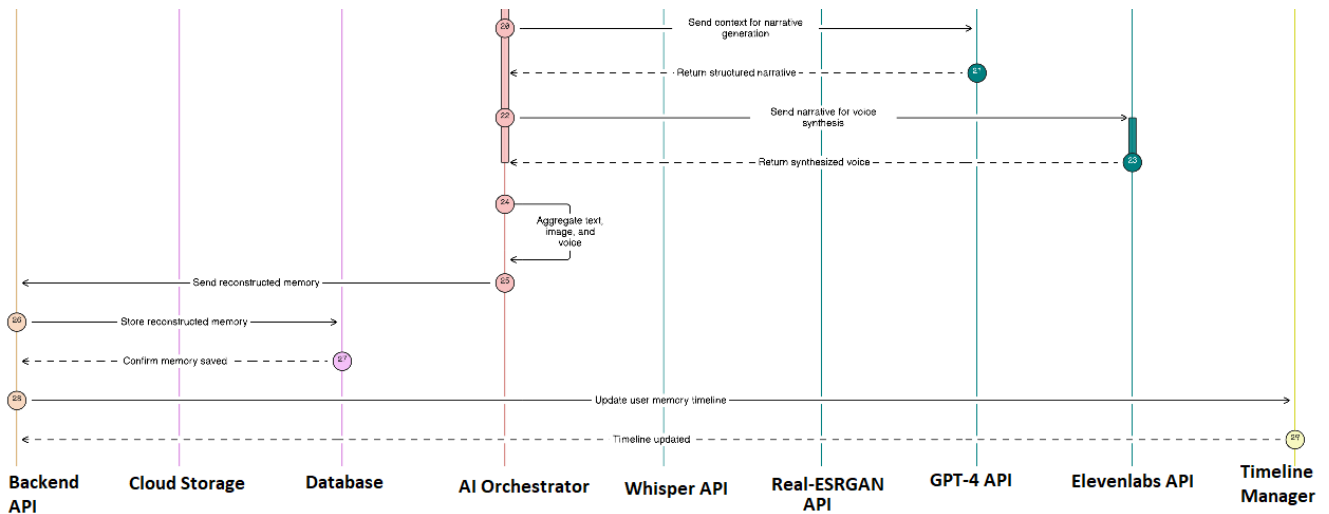
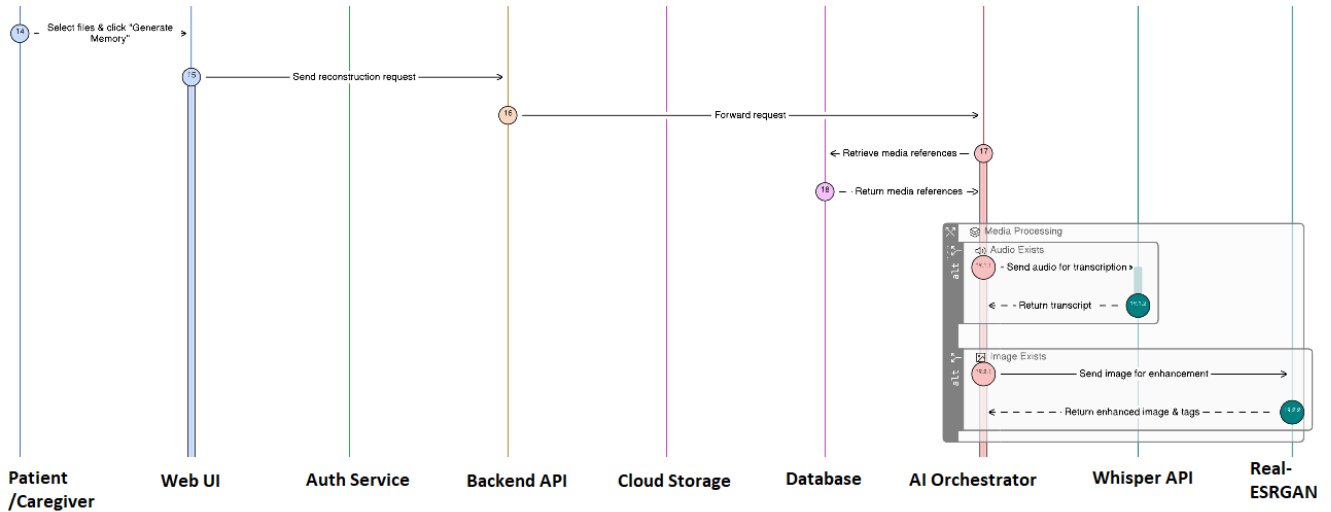
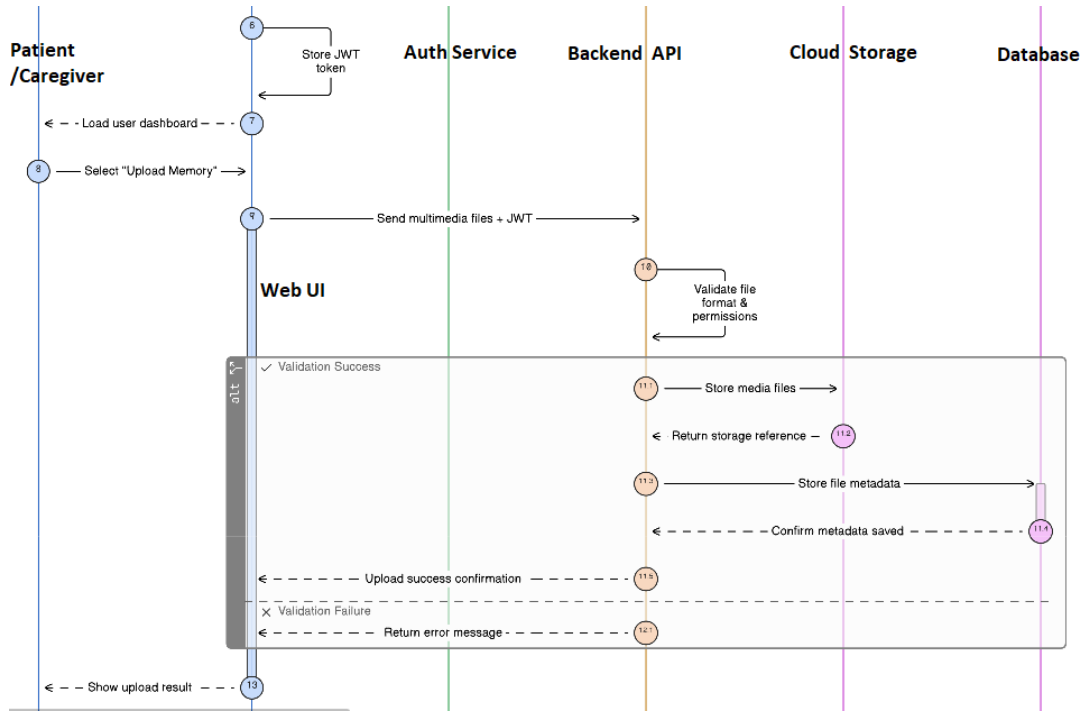
ReMIND: Generative AI Based Memory Rebuilder

influences the separation of user interface, business logic, and data management responsibilities. Additionally, middleware-based authentication and authorization mechanisms are implemented to provide secure access control throughout the system.

Development and collaboration tools such as GitHub, Postman, Figma, and modern package management systems were also reused as part of standard software engineering practices. Overall, ReMIND combines reused technologies with original system design, workflow integration, data modeling, and memory reconstruction logic, resulting in a functional and scalable proof-of-concept platform.



ReMIND: Generative AI Based Memory Rebuilder



ReMIND: Generative AI Based Memory Rebuilder

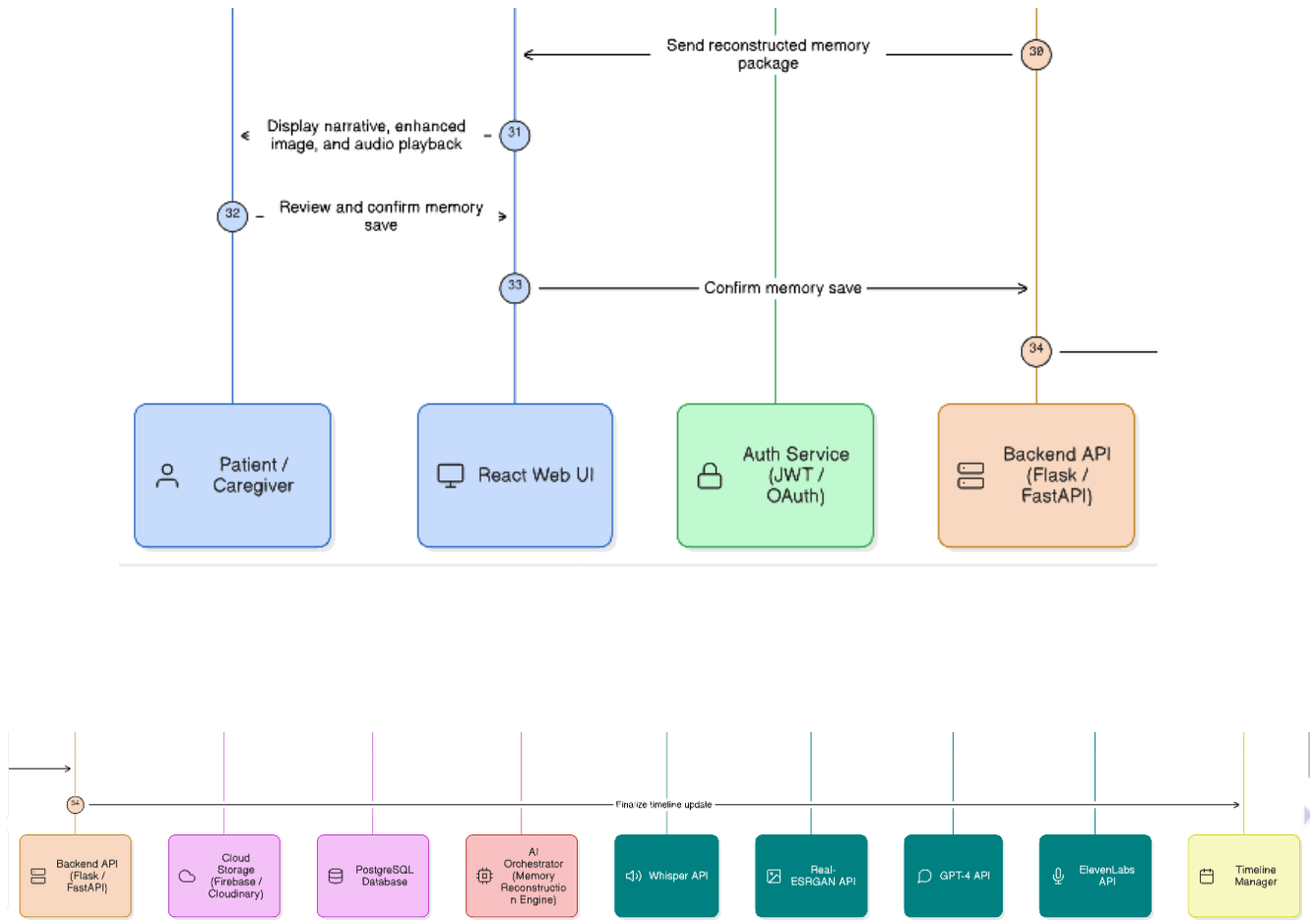


Fig 4.3.1: Sequence Diagram

4.4 Technology Architecture

Data and Processing Characteristics

ReMIND collects, processes, and manages both structured and unstructured data generated during memory reconstruction workflows. The system maintains user credentials, authentication information, role assignments, memory artifacts, reminders, timelines, feedback records, audit logs, and AI-generated outputs. Additionally, the platform manages multimedia inputs including text notes, images, and voice recordings submitted by users for memory reconstruction.

The system operates in an online, event-driven environment where user interactions trigger near real-time processing workflows. When multimedia content is uploaded, specialized AI modules perform contextual analysis, speech processing, image enhancement, and memory reconstruction before generating user-facing outputs. Processing is primarily transactional in nature, while system logs, evaluation of metrics, and feedback records provide basic analytical reporting and monitoring capabilities.

Technology Stack and Platforms:

- **Programming Languages:** Python, JavaScript
- **Frontend Technologies:** React.js, React Router, Context API, Axios, CSS3
- **Backend Technologies:** FastAPI, RESTful APIs, SQLAlchemy, JWT Authentication, OAuth2
- **Text Processing Technologies:** Groq Llama 3, Retrieval-Augmented Generation (RAG), FAISS, Sentence Transformers
- **Voice Processing Technologies:** ElevenLabs STT, PyDub, Librosa, Google Text-to-Speech
- **Image Processing Technologies:** GFPGAN, Real-ESRGAN, Blip
- **Database Platform:** PostgreSQL, Alembic
- **Media Storage Platform:** Cloudinary
- **Version Control and Collaboration:** GitHub
- **API Testing and Validation:** Postman
- **UI/UX Design and Prototyping:** Figma
- **Deployment and Hosting:** Hostinger VPS deployment and hosting.
- **Development Hardware:** Personal laptops and cloud-based computing resources
- **End-User Interface:** Browser-based interface accessible through desktop and mobile devices

Interface and Network Architecture:

ReMIND provides a browser-based user interface accessible through modern web browsers on desktop and mobile devices. No specialized client software is required for system access. The platform follows a WAN-based architecture where users, caregivers, and administrators interact with the system remotely over the Internet.

Communication between the frontend, backend, database, cloud storage, and AI processing services occurs through secure RESTful APIs. This architecture enables seamless integration between system components while maintaining security, reliability, and accessibility for end users regardless of their physical location.

4.5 Architecture Evaluation

Frontend Technology Selection:

Selected Technology: React.js with CSS3

Reason for Selection:

React.js was selected because it provides a component-based architecture that simplifies the development of interactive and responsive user interfaces. Since ReMIND requires real-time user interaction, multimedia uploads, chat-based communication, role-based dashboards, and dynamic content rendering, React offers an efficient and scalable solution. CSS3 was used to create a responsive, modern, and elderly-friendly interface while maintaining full control over styling and accessibility features.

Advantages:

- Component-based architecture promotes code reusability and maintainability.
- Fast and responsive user experience through Virtual DOM rendering.
- Easy integration with RESTful APIs and backend services.
- Supports dynamic content updates without full page reloads.
- Large developer community and extensive ecosystem support.
- Responsive design support for both desktop and mobile devices.

Disadvantages:

- Requires understanding of modern JavaScript concepts and React lifecycle management.
- State management can become complex in large-scale applications.
- Initial project structure and setup require additional configuration compared to traditional HTML-based approaches.

Alternative Considered: Angular

Backend Technology Selection:

Selected Technology: FastAPI

Reason for Selection:

FastAPI was selected because it provides a lightweight, high-performance framework for building RESTful APIs while maintaining excellent compatibility with artificial intelligence and machine learning libraries written in Python. Since ReMIND integrates multiple AI modules, authentication services, database operations, and cloud-based resources, FastAPI enabled rapid development, clean architecture, and efficient communication between system components.

Advantages:

- High-performance asynchronous request handling.
- Seamless integration with AI and data processing libraries.
- Automatic API documentation generation.
- Lightweight and modular architecture.
- Strong support for RESTful API development.
- Easy integration with authentication, database, and cloud services.

Disadvantages:

- Fewer built-in features compared to larger frameworks.
- Requires manual configuration of some components.
- Smaller ecosystem compared to older Python frameworks.

Alternative Considered: Flask

AI Models and API Selection:

The ReMIND system utilizes specialized AI models and APIs for text processing, voice processing, and image enhancement. Instead of training custom machine learning models, the project integrates existing AI technologies to achieve high-quality results while remaining technically and computationally feasible

Groq Llama 3 (Text Processing and Memory Reconstruction):

Reason for Selection:

Groq Llama 3 was selected because it provides fast and context-aware text generation capabilities suitable for memory reconstruction. It enables the system to understand fragmented user inputs, generate coherent narratives, and support Retrieval-Augmented Generation (RAG) workflows while maintaining low response latency.

Advantages:

- Fast inference and response generation.
- Strong contextual understanding and reasoning capabilities are important.
- Suitable for conversational memory reconstruction.
- Easy integration with RAG pipelines.
- Open-Source Availability.

Disadvantages:

- Depends on external API availability.
- Generated responses may occasionally require validation.
- Performance is influenced by prompt quality and retrieved context.

Alternative Considered: OpenAI GPT-4

ElevenLabs Speech-to-Text (STT):

Reason for Selection:

ElevenLabs STT was selected because it provides accurate speech recognition and transcription capabilities across different accents and speaking styles. This is particularly important for elderly users who may have varied speech patterns and audio quality.

Advantages:

- High transcription accuracy.
- Handles different accents effectively.
- Supports natural conversational speech.
- Easy API integration.
- Reliable performance on voice recordings.

Disadvantages:

- Requires internet connectivity.
- API usage costs may increase with scale.
- Depends on third-party service availability.

Alternative Considered: OpenAI Whisper

Google Text-to-Speech (TTS):

Reason for Selection:

Google Text-to-Speech was selected because it offers reliable voice synthesis with simple integration and low computational requirements. It enables reconstructed memories to be delivered as audio output, improving accessibility, and usability for users.

Advantages:

- Simple and reliable implementation.

- Natural-sounding speech output.
- Supports multiple languages and voices.
- Lightweight, Open-Source, and efficient.
- Well-documented API support.

Disadvantages:

- Limited emotional expressiveness compared to premium voice services.
- Requires internet connectivity.
- Voice customization options are limited.

Alternative Considered: ElevenLabs TTS

Real-ESRGAN, GFPGAN, and BLIP (Image Processing):

Reason for Selection:

These models were selected to improve the quality and usefulness of historical photographs used during memory reconstruction. Real-ESRGAN enhances image resolution, GFPGAN restores facial details, and BLIP generates image descriptions that contribute contextual information for memory generation.

Advantages:

- High-quality image enhancement and restoration.
- Effective recovery of degraded or low-resolution photographs.
- Automatic caption generation for contextual understanding.
- Improves quality of multimedia inputs.
- Open-source and research-supported technologies.

Disadvantages:

- Computationally intensive processing.
- Processing time increases for large images.
- Caption generation accuracy depends on image quality.

Alternative Considered: OpenCV-based image enhancement techniques

Database Selection:

Selected Technology: PostgreSQL with Alembic Migrations

Reason for Selection:

PostgreSQL was selected because it provides a reliable, scalable, and feature-rich relational database management system capable of handling the structured data requirements of ReMIND. The platform stores user accounts, authentication information, memory artifacts, reminders, timelines, feedback records, audit logs, and system metadata. Alembic was selected to manage database schema migrations and version control, ensuring consistent database updates throughout development and deployment.

Advantages:

- Open-source and highly reliable.
- Strong support for relational data modeling is important.
- Excellent data integrity and consistency.
- Advanced querying and indexing capabilities.
- Scalable for future system growth.
- Alembic enables controlled and versioned database migrations.

Disadvantages:

- Requires proper schema design and maintenance.
- More complex than lightweight database solutions.
- Migration management requires careful version control.

Alternative Considered: MongoDB

Media Storage Selection:

Selected Technology: Cloudinary

Reason for Selection:

Cloudinary was selected as the primary media storage platform because it provides secure cloud-based storage, efficient media delivery, and seamless integration with web applications. Since ReMIND processes and stores user-uploaded images for memory reconstruction, Cloudinary offered a reliable solution for media management while reducing the complexity of maintaining dedicated file storage infrastructure.

Advantages

- Secure cloud-based image storage.
- Fast media delivery through CDN support.
- Easy integration with backend services and APIs.
- Automatic image optimization and transformation features.
- Reduces server storage requirements.
- Scalable storage and media management capabilities.

Disadvantages

- Storage and bandwidth limit free plans.
- Dependence on third-party service availability.
- Additional costs may arise as storage requirements increase.

Alternative Considered: Firebase Storage

Deployment Infrastructure Selection:

Selected Technology: Hostinger VPS

Reason for Selection:

Hostinger VPS was selected because it provides a dedicated and cost-effective hosting environment with full server control, allowing the complete ReMIND system to be deployed on a single infrastructure. The platform supports backend services, database connectivity, API integrations, and web hosting requirements while offering sufficient performance for a proof-of-concept AI-assisted application. The VPS environment also provides flexibility for configuration, deployment, monitoring, and future scalability.

Advantages:

- Full administrative control over server resources.
- Supports deployment of complete application stacks on a single platform.
- Cost-effective compared to many enterprise cloud solutions.
- Dedicated computing resources improve reliability and performance.
- Flexible environment for custom configurations and software installation.

Disadvantages:

- Requires manual server configuration and maintenance.
- Security updates and system monitoring are the responsibility of administrators.
- Limited resources compared to large-scale cloud infrastructure platforms.
- Downtime recovery and backup management require additional configuration.

Alternative Considered: Render

Chapter 5. Detailed Design and Implementation

5.1 Component-Component Interface

The ReMIND system follows a modular architecture in which internal components interact through well-defined service interfaces and RESTful APIs. The Frontend Layer communicates with the Backend API for user authentication, memory submission, memory retrieval, profile management, and administrative operations. The Backend Layer acts as the central controller and coordinates requests between the database, cloud storage, and AI processing modules.

The AI Orchestration component manages communication with the Text Processing, Voice Processing, and Image Processing modules. Each module performs specialized processing tasks and returns structured outputs that are combined to generate reconstructed memories. The Data Layer stores user information, memory artifacts, audit logs, and system metadata, while media files are maintained through cloud-based storage services.

This separation of responsibilities ensures loose coupling, maintainability, scalability, and efficient communication between software components while simplifying future enhancements and system integration.

Deployment Diagram:

The Deployment Diagram illustrates the physical architecture of the ReMIND system and the interaction between its major components. Users access the platform through a web browser, where the React.js frontend provides interfaces for memory submission, retrieval, and system interaction. All user requests are securely transmitted via HTTPS to the backend server, which is responsible for authentication, application logic, data processing, and coordination of system services. The backend integrates with several AI-powered services, including GPT-4 for memory narrative generation, Whisper for speech transcription, ElevenLabs for voice synthesis, and Real-ESRGAN for image enhancement. User accounts, memory records, and other structured information are stored in a PostgreSQL database, while multimedia content such as images, audio files, and generated artifacts are maintained in cloud storage services. After processing and reconstruction are completed, the backend delivers the generated memories, enhanced media, and relevant notifications back to the frontend for presentation to the user. This deployment architecture enables the ReMIND system to operate as a scalable, secure, and cloud-based platform powered by advanced artificial intelligence technologies.

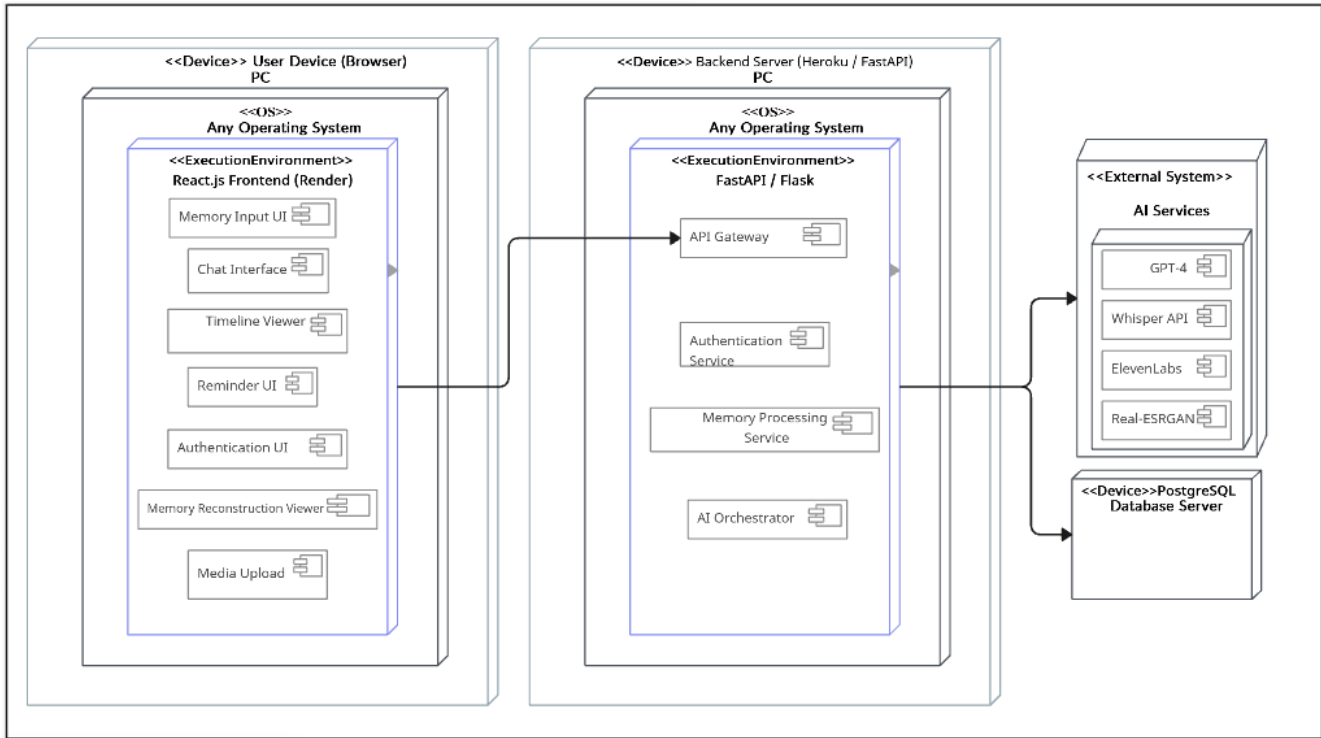


Fig 5.1.1: Deployment Diagram

5.2 Component-External Entities Interface

The ReMIND system interacts with multiple external services to provide memory reconstruction functionality. The Backend API serves as the central integration point between internal system components and third-party services. External AI technologies are utilized for text understanding, memory generation, speech recognition, voice synthesis, image enhancement, and image caption generation.

The platform communicates with Groq Llama 3 for contextual memory reconstruction, ElevenLabs for speech-to-text processing, Google Text-to-Speech for audio generation, GFPGAN and Real-ESRGAN for image restoration and enhancement, and BLIP for image caption generation. Cloudinary is used for cloud-based media storage, while PostgreSQL provides persistent storage for structured application data.

Communication with external services is performed through secure RESTful APIs and internet-based protocols. This integration approach enables the system to leverage advanced AI capabilities without requiring the development and maintenance of custom machine learning models.

5.3 Component-Human Interface

The human interface of ReMIND is designed to provide an accessible, intuitive, and user-friendly experience for individuals with varying levels of technical expertise. The system includes interfaces for authentication, profile setup, memory submission, chat-based interaction, memory history viewing, administrative management, and system monitoring.

Users interact with the platform through a browser-based interface that supports text, image, and voice inputs. The design emphasizes accessibility through simplified navigation, readable typography, consistent layouts, responsive design principles, and clearly labeled controls. Special consideration has been given to elderly users and individuals experiencing cognitive challenges by minimizing complexity and reducing cognitive load.

Administrative users are provided with dedicated dashboards for monitoring system activities, managing users, reviewing analytics, and accessing audit logs. Overall, the interface promotes ease of use while maintaining functionality, accessibility, and security throughout the user experience.

5.4 Screenshots/Prototype

5.4.1 Workflow

The workflow of ReMIND begins when a user accesses the platform and completes authentication. Following successful login, the user can provide memory-related information through text, image, or audio inputs. The submitted content is validated and forwarded to the appropriate AI processing modules for analysis, enhancement, transcription, caption generation, and contextual understanding.

The generated outputs are aggregated by the memory reconstruction engine, which produces a coherent memory narrative. The reconstructed memory is then stored together with associated metadata and becomes available for future retrieval. Users can access previously generated memories, review historical records, and continue interacting with the system through the conversational interface.

Administrative processes operate concurrently to support monitoring, audit logging, user management, and system maintenance. The workflow follows an event-driven architecture that ensures efficient coordination between users, application services, databases, cloud storage platforms, and AI processing technologies.

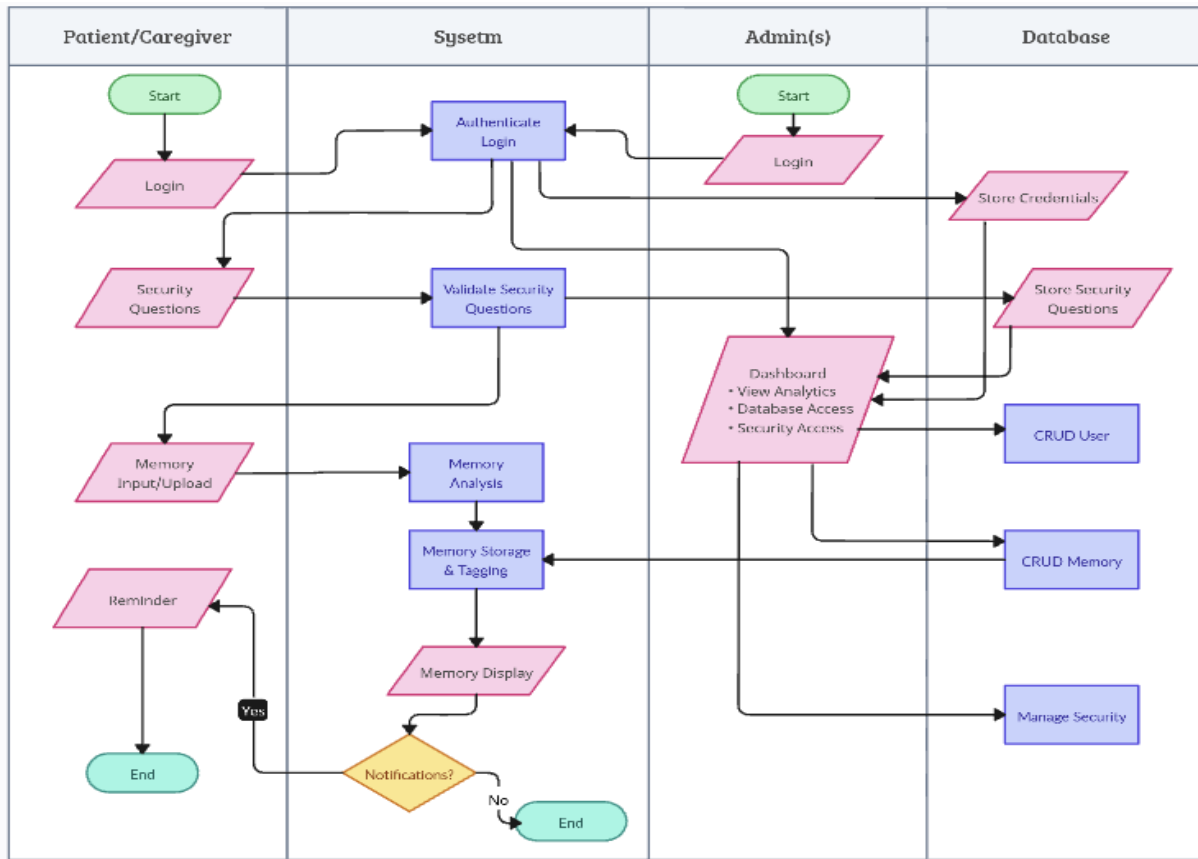


Fig 5.4.1: Swimlane Workflow Diagram

5.4.2 Screens

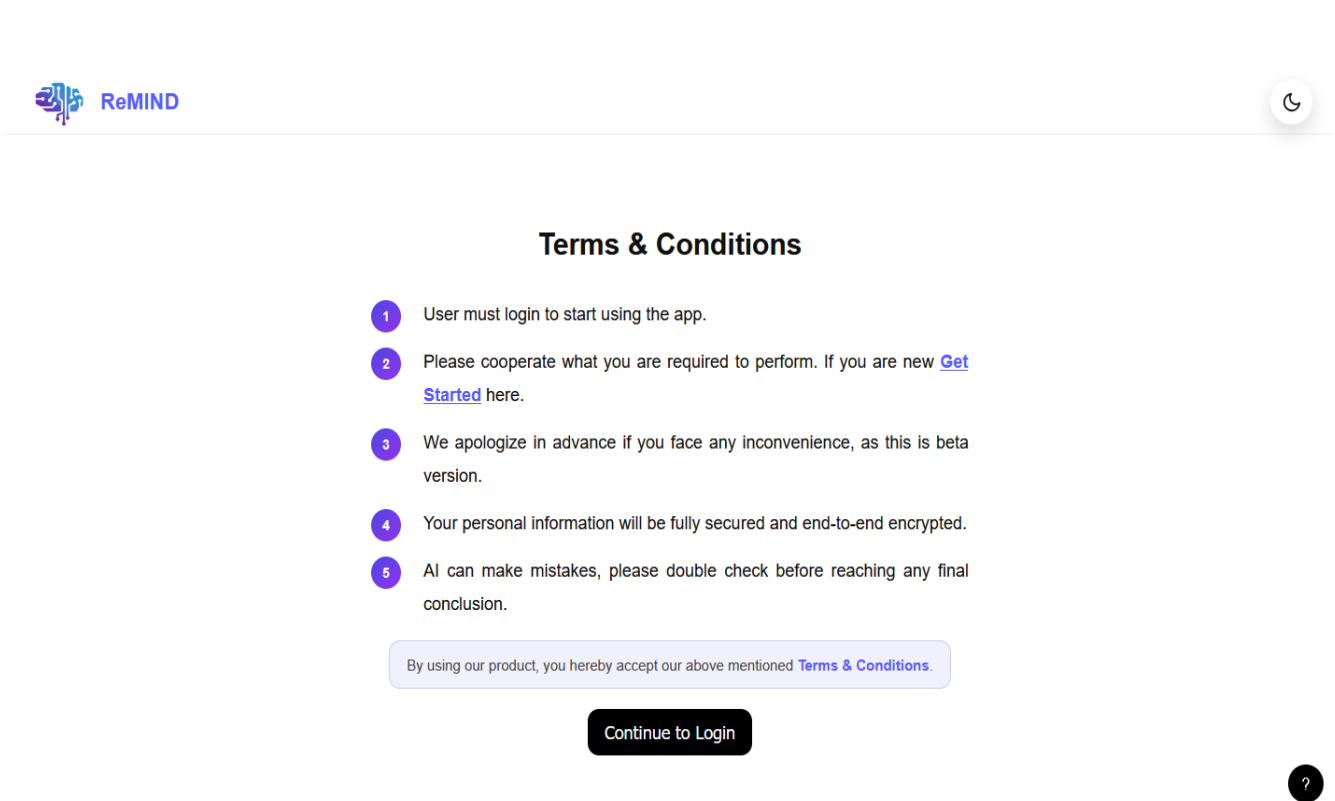


Fig 5.4.2.1: Terms & Condition Page

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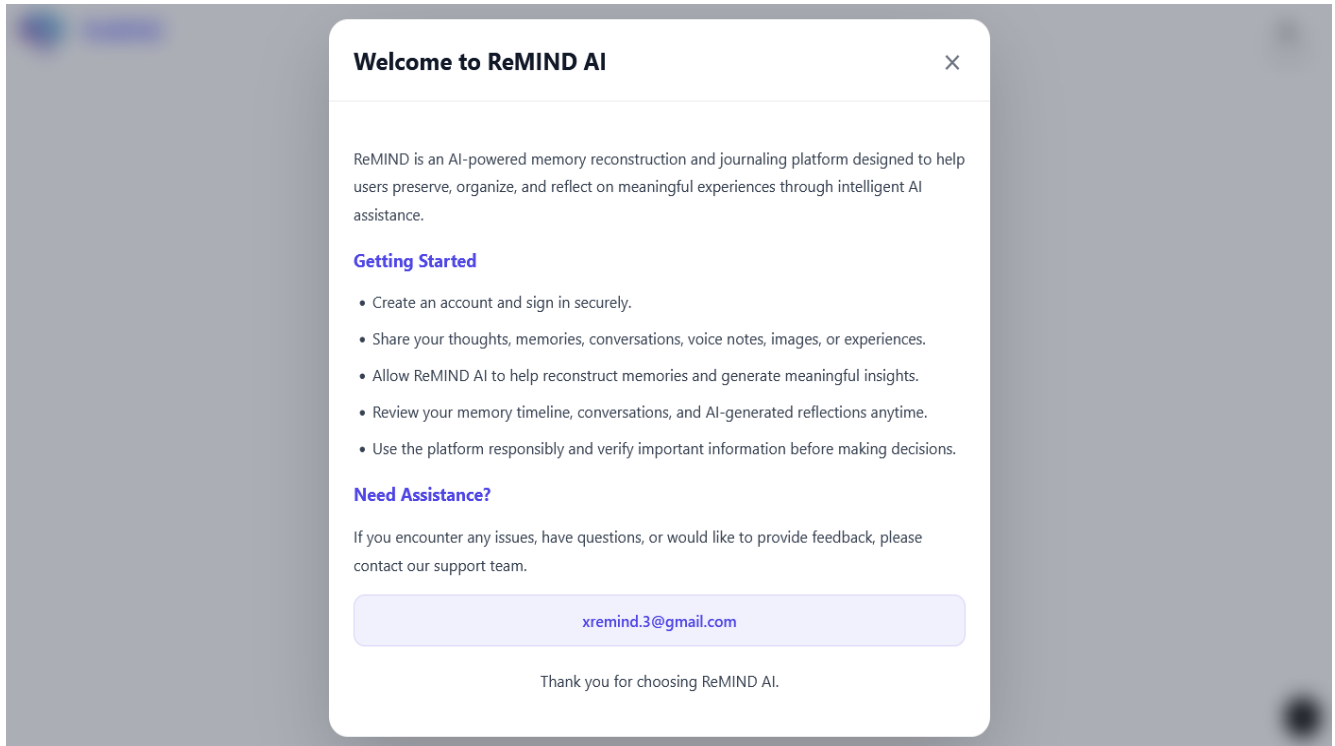


Fig 5.4.2.2: Help Popup

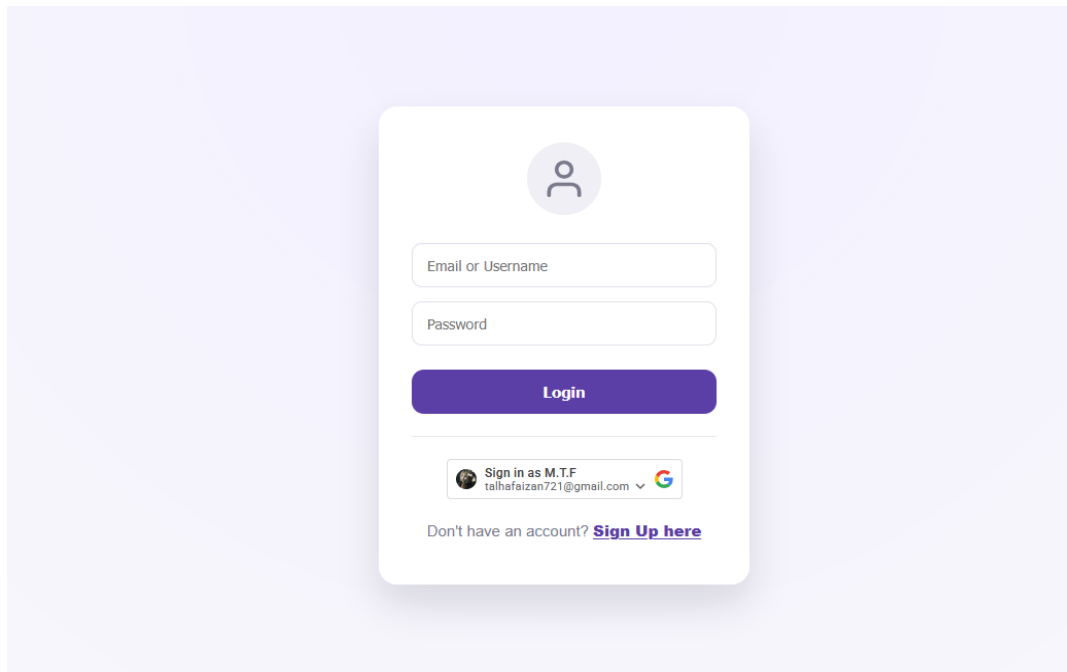
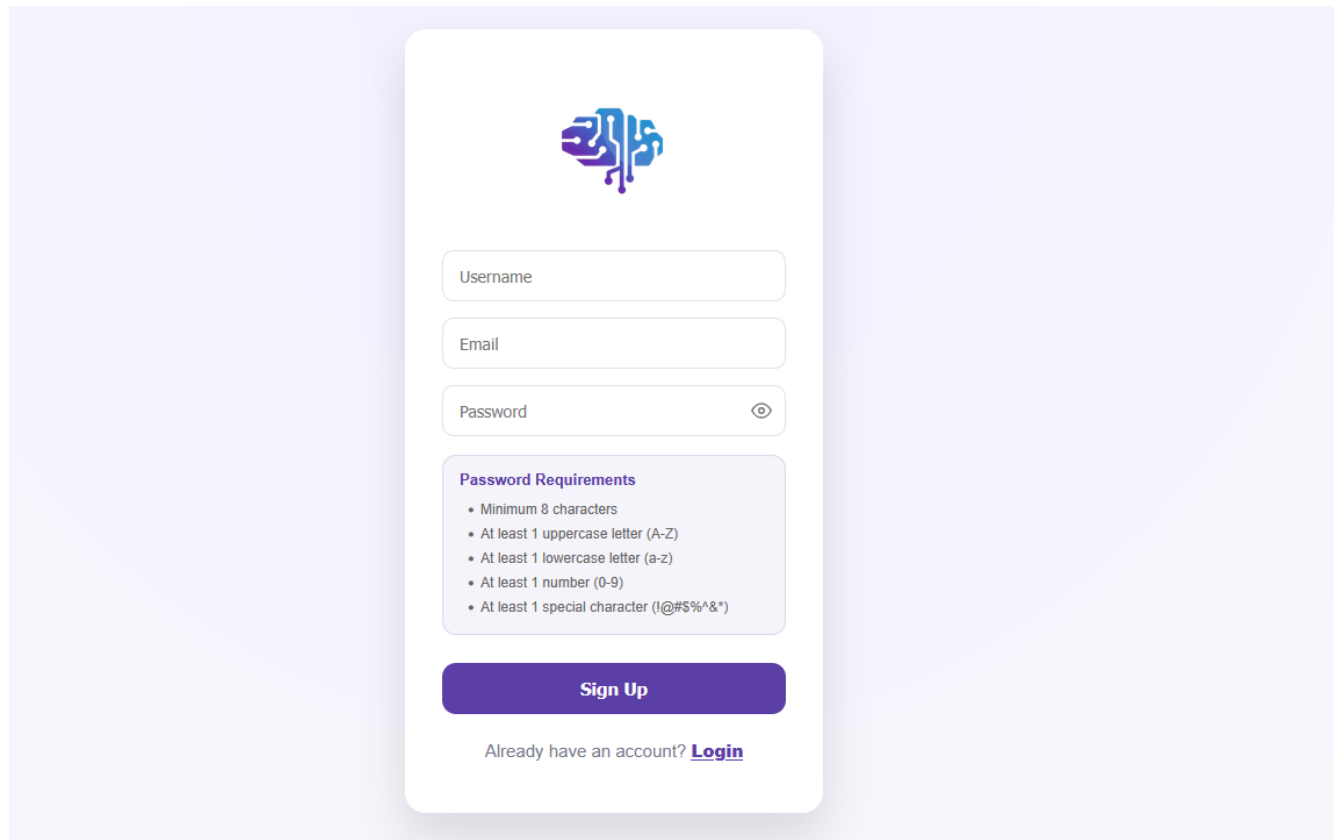


Fig 5.4.2.3: Login Page

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The sign-up page features a central white card on a light purple background. At the top of the card is a stylized brain logo with circuit-like patterns in blue and purple. Below the logo are three input fields: 'Username', 'Email', and 'Password'. The 'Password' field includes an eye icon for toggling visibility. Under the password field is a section titled 'Password Requirements' with a bulleted list: 'Minimum 8 characters', 'At least 1 uppercase letter (A-Z)', 'At least 1 lowercase letter (a-z)', 'At least 1 number (0-9)', and 'At least 1 special character (!@#\$\$%^&*)'. A large purple 'Sign Up' button is positioned below the requirements. At the bottom of the card, it says 'Already have an account? [Login](#)'.

Fig 5.4.2.4: Sign Up Page

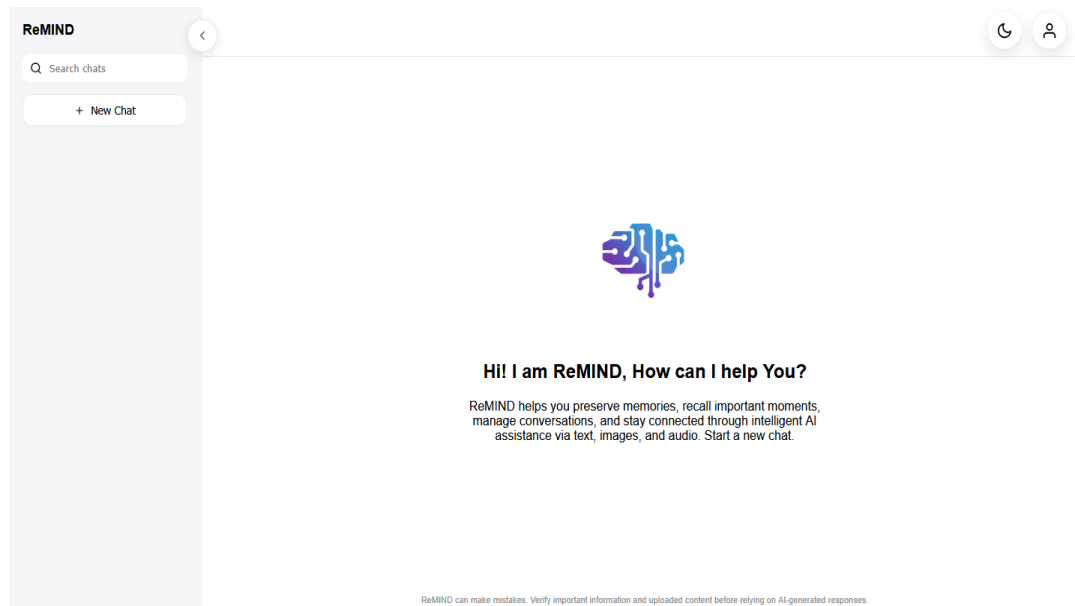


Fig 5.4.2.5: Main Chatting UI

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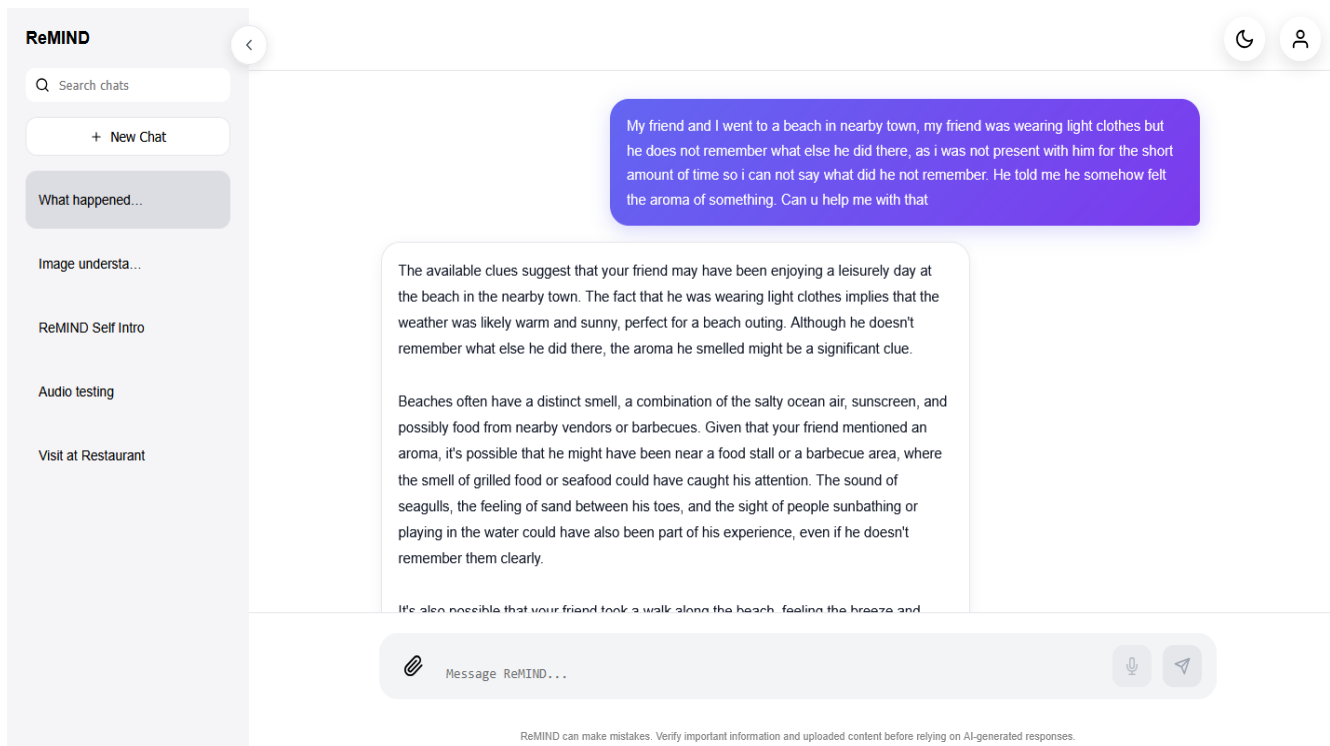


Fig 5.4.2.6: Sample Functional Chat

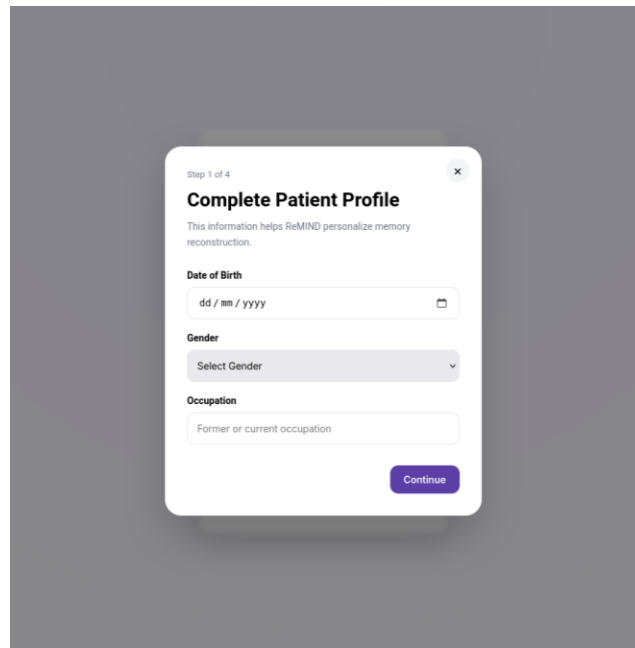


Fig 5.4.2.7: Patient Profile Popup

Chapter 6. Test Specification and Results

6.1 Test Case Specification

Identifier	TC-1
Related requirements(s)	User Authentication Module
Short description	Verify that a registered user can log into the system successfully.
Pre-condition(s)	User account exists.
Input data	Valid username and password
Detailed steps	Open login page → Enter credentials → Click Login
Expected result(s)	User authenticated and redirected to the dashboard.
Post-condition(s)	JWT session created.
Actual result(s)	User successfully authenticated and loaded dashboard.
Test Case Result	PASS

Table 6.1.1: User Authentication (Login)

Identifier	TC-2
Related requirements(s)	User Authentication Module
Short description	Verify successful user registration.
Pre-condition(s)	Username and Email do not already exist.
Input data	New username, email, password
Detailed steps	Open registration page → Enter details → Submit
Expected result(s)	The user account was created successfully.
Post-condition(s)	User records stored in PostgreSQL.
Actual result(s)	The user account was created successfully.
Test Case Result	PASS

Table 6.1.2: User Registration

Identifier	TC-3
Related requirements(s)	Memory Reconstruction Module
Short description	Verify memory generation from text input.
Pre-condition(s)	The User logged in.
Input data	Textual memory fragment
Detailed steps	Enter text → Submit

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Expected result(s)	AI-generated memory narrative returned.
Post-condition(s)	Memory is stored in the database.
Actual result(s)	The memory was reconstructed successfully.
Test Case Result	PASS

Table 6.1.3: Text Memory Reconstruction

Identifier	TC-4
Related requirements(s)	Multimedia Input Module
Short description	Verify image upload and processing workflow.
Pre-condition(s)	The User Logged in.
Input data	Valid Format Image
Detailed steps	Upload image → Submit
Expected result(s)	Image processed and metadata generated.
Post-condition(s)	Image stored in Cloudinary.
Actual result(s)	Image processed successfully.
Test Case Result	PASS

Table 6.1.4: Image Upload and Processing

Identifier	TC-5
Related requirements(s)	Voice Processing Module
Short description	Verify audio upload and speech processing.
Pre-condition(s)	The User logged in.
Input data	Valid Format Audio File
Detailed steps	Upload Audio -> submit
Expected result(s)	Speech converted to text and processed.
Post-condition(s)	Voice data was processed successfully.
Actual result(s)	Audio transcription was generated successfully.
Test Case Result	PASS

Table 6.1.5: Voice Upload and Processing

Identifier	TC-6
Related requirements(s)	Memory Management Module
Short description	Verify retrieval of previously generated memories.
Pre-condition(s)	Existing memories are available.
Input data	User requests.
Detailed steps	Open memory history page.
Expected result(s)	Previously generated memories were displayed.

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Post-condition(s)	Memory records were retrieved successfully.
Actual result(s)	Memory history loaded correctly.
Test Case Result	PASS

Table 6.1.6: Memory History Retrieval

Identifier	TC-7
Related requirements(s)	Application Architecture
Short description	Verify successful API communication.
Pre-condition(s)	Backend running.
Input data	API request.
Detailed steps	Send request → Receive response
Expected result(s)	HTTP 200 responses were returned.
Post-condition(s)	Data exchanged successfully.
Actual result(s)	Successful communication is established.
Test Case Result	PASS

Table 5.1.7: Frontend Backend Communication

Identifier	TC-8
Related requirements(s)	Database Management Module
Short description	Verify data persistence and retrieval.
Pre-condition(s)	Database is available.
Input data	User and memory records.
Detailed steps	Insert → Retrieve → Verify
Expected result(s)	Data stored and retrieved correctly.
Post-condition(s)	Database consistency maintained.
Actual result(s)	Database operations are executed successfully.
Test Case Result	PASS

Table 6.1.8: Database Operations

Identifier	TC-9
Related requirements(s)	Media Storage Module
Short description	Verify image storage in Cloudinary.
Pre-condition(s)	Cloudinary configured.
Input data	Image file
Detailed steps	Upload image
Expected result(s)	File stored and URL generated.
Post-condition(s)	Media accessible through Cloudinary.

Actual result(s)	Image stored successfully.
Test Case Result	PASS

Table 6.1.9: Cloudinary Media Storage

Identifier	TC-10
Related requirements(s)	End-to-End System Integration
Short description	Verify complete system workflow from input to memory generation.
Pre-condition(s)	The user logged in.
Input data	Text, image, and voice inputs.
Detailed steps	Upload inputs → Process → Generate memory
Expected result(s)	Unified reconstructed memory generated.
Post-condition(s)	Memory stored and displayed.
Actual result(s)	The end-to-end workflow was completed successfully.
Test Case Result	PASS

Table 6.1.10: Complete Memory Reconstruction Workflow

6.2 Summary of Test Results

Module Name	Test cases run	Number of defects found	Number of defects corrected so far	Number of defects still need to be corrected
Authentication	TC1, TC2	3	3	0
Memory Reconstruction	TC3, TC10	6	6	0
Image Processing	TC4	4	4	0
Audio Processing	TC5	5	5	0
Memory Management	TC6	1	1	0
API Communication	TC7	8	8	0
Database Management	TC8	6	6	0
Media Storage	TC9	1	1	0
Complete System	TC1 - TC10	34	34	0

Table 6.2: Summary of Test Results

Chapter 7. Project Completion Status/Conclusion

Module Name	Status
User Authentication Module (Login / Signup / Session Management)	Complete
User Interface Module (React Frontend UI, Navigation, Dashboard)	Complete
Memory Input Module (Text, Image, and Audio)	Complete
Multimedia Processing Module (Image Enhancement, Face Restoration, Image Captioning)	Complete
Speech Processing Module (Speech-to-Text, Audio Processing, Voice Synthesis)	Complete
Memory Reconstruction Module (Context-Aware AI Narrative Generation)	Complete
Backend API Module (FastAPI Services and System Integration)	Complete
Database Management Module (PostgreSQL, SQL Alchemy, Alembic Migrations)	Complete
Cloud Media Storage Module (Cloudinary Integration)	Complete
Administration and Security Module (RBAC, Audit Logs, Protected Routes)	Complete
Testing and Validation Module	Complete
Event Clustering and Timeline Management Module	Partially Implemented
Reminder and Notification Module	Not Complete
Complete System	Complete

Table 7.1: Project Completion Status

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Target/Objective	Status	Reason(s)
Memory Reconstruction	Completed	
Reminder and Notifications	Not Completed	Due to project scope and time constraints, reminder scheduling and notification functionality were not implemented in the final deployed system.
Elderly Friendly Interface	Completed	
Secure Data Handling	Completed	
Proof of Concept Demonstration	Completed	
Caregiver Collaboration	Partially Completed	Anyone on behalf of a patient from user account can access and use the system, but specific separate caregiver role-based integration was not fully implemented due to time constraints.
Number of Targets Completed	4/6	
Number of Targets Partially Completed	1/6	
Number of Targets Not Completed	1/6	

Table 7.2: Objectives/Target completion status.

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Appendix A Glossary

Appendix B IV & V Report

(Independent verification & validation)

IV & V Resource

Name

Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table B.1: List of non-trivial defects

Appendix C Deployment/Installation Guide

Accessing the Deployed System:

ReMIND is currently deployed and publicly accessible through its production URL:

<https://remindio.space>

Users can access the platform directly through a modern web browser without requiring any local installation. The deployed version of the system is scheduled to remain available until **1 July 2026**.

System Requirements:

- Modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari)
- Stable internet connection
- Microphone access for voice-based features
- Device capable of uploading image and audio files

Local Installation (After Deployment Period):

After the public deployment period ends, the project can be accessed and installed locally through the official GitHub repository.

Prerequisites:

- Python 3.10 or later
- Node.js and npm
- PostgreSQL and Git
- Internet connection
- Required API credentials

Backend Setup:

- Clone the project repository.
- Navigate to the backend directory.
- Create and activate a Python virtual environment.
- Install project dependencies.
- Configure environment variables and API credentials.
- Apply database migrations using Alembic.
- Start the FastAPI server.

Frontend Setup:

- Navigate to the frontend directory.
- Install required npm packages.
- Configure environment variables if required.
- Start the React application.

Database Setup:

1. Install PostgreSQL.
2. Create the required database.
3. Configure database connection settings.
4. Execute database migrations.

Verification:

After successful installation:

- The frontend interface should be loaded correctly.
- User registration and authentication should function properly.
- Text, image, and voice reconstruction features should operate successfully.
- Database connectivity should be established.
- AI services should generate memory reconstruction outputs correctly.

Appendix D User Manual

Introduction:

ReMIND is an AI-assisted memory reconstruction platform that helps users reconstruct meaningful memories from fragmented text, image, and voice inputs. The system combines multiple AI technologies to generate context-aware memory narratives through a unified web interface.

Registration and Login:

- Open the ReMIND web application.
- Register a new account or sign in using existing credentials.
- Alternatively, use Google Authentication if available.
- Upon successful authentication, the system redirects the user to the main dashboard.

Text-Based Memory Reconstruction:

- Navigate to the memory reconstruction interface.
- Enter a memory fragment, note, description, or contextual information.
- Submit the input for processing.
- Review the generated memory narrative returned by the system.

Image-Based Memory Reconstruction:

- Select the image upload option.
- Upload an image from your device.
- Wait for image enhancement and analysis.
- Review the generated visual description and reconstructed memory output.

Voice-Based Memory Reconstruction:

- Select the voice input option.
- Record audio or upload a supported audio file.
- Submit the recording for processing.
- Review the generated transcription and reconstructed memory narrative.

Viewing Generated Memories:

- Access to the memory history section.
- Browse previously generated memory reconstructions.
- Select a memory to view its complete details and associated outputs.

Administration Features:

Authorized administrators can:

- View registered users.
- Monitor system activity.
- Access platform analytics.
- Review audit logs.
- Manage system records.

Logout:

- Open the navigation menu.
- Select the Logout option.
- The system will terminate the active session and redirect the user to the login page.

Troubleshooting:

Unable to Upload Media:

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- Verify internet connectivity.
- Ensure supported file formats are being used.
- Confirm that uploaded files are not corrupted.

Voice Features Not Working:

- Verify microphone permissions.
- Check the browser audio settings.
- Ensure a stable internet connection.

Account Access Issues:

- Verify the entered username/email and password.
- If account information needs to be updated or access issues persist, contact the ReMIND administrators.
- Users may also refer to the project's GitHub repository or contact the development team through the official project email for assistance.

System Unavailable:

The public deployment is scheduled to remain available until **1 July 2026**. After this period, users can access the source code through the GitHub repository and deploy the system locally following the installation guide provided above.